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Editorial

Why another journal, and why this one

Dr. Ravi Kumar, Editor in Chief

There are a great many journals. The honest question for anyone starting a new one is what this journal will do that the existing ones do not.

The answer is not novelty of subject. Law, nursing, literature, history, education, policy and environmental science are all well served by excellent journals, and this journal does not pretend to compete with them at the top of their fields. The answer is a matter of access, and of how a manuscript is treated when it arrives.

Consider the ordinary situation of a college teacher in a district town who has spent two years studying something worth knowing: how cases move through the courts, how nursing students cope with the demands of training, what a forgotten Hindi periodical reveals about its readership. That research is often good. What happens to it is often one of three things. It goes to a journal that publishes anything for a fee, where nobody reads it and nobody cites it. It goes to a journal that never replies. Or it is not written up at all, because the author reasonably concludes that neither outcome is worth two years of work.

This journal exists for that manuscript. Three commitments follow from it, and they are commitments rather than aspirations because a reader can check each one.

The first is that we reply. Every submission receives a written editorial response within seven working days, and every manuscript that goes to review receives the reviewers' reports, whether the decision is acceptance or rejection.

The second is that we charge nothing. No submission fee, no processing charge, no publication charge, no fee for a certificate, no paid fast track. A publication charge decides who can publish before anyone has read a word.

The third is that the standard does not move. Review is double anonymous, against criteria published on this website. Those criteria apply equally to a manuscript from a national institute and a manuscript from a college nobody has heard of.

In this issue

This founding issue brings together four articles from law and the health sciences. The first offers a five-year empirical analysis of case pendency in the High Courts of India and argues for measures that would make adjudication faster without making it less fair. Two studies from nursing education follow: one measures the level of stress among B.Sc. Nursing students in Guwahati, Assam, and the other assesses their knowledge, skills and practice regarding patient safety. The issue closes with a review of neuropsychiatric disorders, their causes, diagnosis and management.

We are not indexed yet, and we say so plainly on the indexing and preservation page rather than implying accreditation we do not hold. We do not have an impact factor, and we will not display a purchased one. We are new, and the only honest response to that is to do the work well from the first issue and let the record accumulate.

Manuscripts are now invited for Volume 1, Issue 2 (October to December 2026). The submission guidelines set out precisely what to send and what happens after you send it.

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RESEARCH ARTICLE · LAW AND CONSTITUTIONAL STUDIES

A QUINQUENNIAL EMPIRICAL ANALYSIS OF CASE PENDENCY IN INDIAN HIGH COURTS AND THE IMPERATIVE FOR EXPEDITED ADJUDICATION

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KEYWORDS

Case Pendency, High Courts, Expedited Adjudication, Judicial Delay, Speedy Justice, Judicial Reforms

ABSTRACT

The growing pendency of cases in the different High Courts of India represents a major obstacle in the timely delivery of justice. In this regard, this paper tries to analyse the problem of case pendency in the Indian High Courts on a quinquennial basis in an attempt to identify the trends in case dispositions and provide for the need for faster resolution of cases. In terms of the data used for the analysis, the paper makes use of secondary data concerning case pendency in the Indian High Courts during the last five years.

It is necessary to note that there is a need for faster case dispositions as a means to deal with several problems, such as an increasing caseload, vacancies, and lack of adequate staff in the higher judiciary. In particular, case pendency results in higher cost and delay for the litigants, prevents quick realisation of rights, and creates doubts about the ability of the court to provide fair justice. Moreover, in the context of criminal cases, the prolonged trial leads to the violation of the fundamental rights of the accused or victims.

The analysis conducted in this paper shows that in order to resolve the problem of case pendency, it is necessary to apply a variety of measures, which will enable faster disposal of cases. First of all, it is necessary to strengthen the higher judiciary, deal with the problem of vacancies, improve the infrastructure, apply such technologies as case-flow management, minimise adjournments, e-courts, virtual courts, and promote mediation and other alternative dispute resolution mechanisms. Finally, the analysis of the list of pending cases on a regular basis could help to dispose of the cases faster.

It is possible to conclude that faster adjudication is crucial for the proper delivery of justice. In its turn, the analysis of the last five years gives important insights into adjudication trends, which could help to resolve pending cases faster.

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Introduction

The Indian judicial system is based on the principle of providing justice, equality and protection of legal rights to every individual. The High Courts play a crucial role in the judicial system of India. They are constitutional courts, which means that they are established by the Constitution of India under Articles 214 to 231. The High Courts have original, appellate, writ, supervisory and administrative jurisdictions over the state. As such, they have a huge load of pending cases and bear heavy responsibility.

The pendency of cases has become a matter of great concern as it not only affects the poor and vulnerable sections of society but also the accused of serious crimes. Further, there are economic implications of the piled-up cases in the High Courts. Hence, the High Courts need to address the issue of pending cases in various matters.

The issue of pendency of cases in the High Courts not only constitutes a major challenge before the judiciary but also has adverse repercussions on the rights of citizens and the overall economy of the country. A significant number of cases remain pending due to vacancies in the judiciary, a rising number of litigations, lack of court infrastructure, shortage of staff, lengthy procedures, frequent adjournments and poor case management. Further, the use of the latest technologies for faster case disposal, like e-filing, video conferencing, e-courts and online tracking of

cases, needs to be facilitated, so as to resolve the problem of pending cases in the judiciary. The study aims to carry out a quinquennial study on the pendency of cases in the High Courts in order to understand the pattern of rise and fall of cases pending in the various High Courts over the years. The study also compares the trend of pendency of cases in different High Courts.

The problem of case pendency in High Courts needs to be addressed by filling up the vacancies in a fast-track manner, proper case-flow management, avoiding frequent adjournments, proper infrastructure and modern technologies to dispose of the cases in time. The faster disposal of pending cases can also be achieved by exploring alternate dispute resolution mechanisms and faster adjudication of cases. However, it is also essential that the rights and interests of the parties involved are not compromised in order to address the pendency of cases in High Courts. The Indian judiciary needs to move towards a faster, effective, transparent and efficient justice delivery system and help build public confidence in the judiciary.

Problem statement

The pendency of cases before the High Courts is a matter of great concern as it deters the very justice delivery mechanism. The issue of pendency, despite numerous judicial reforms, technological advancements and other steps taken to expedite the process, still continues to haunt the judicial system in the country. A large number of cases pending before the judicial authorities puts tremendous strain on the system as well as on the rights of citizens. The problem of pendency before the High Courts is of a different nature as it deals with matters concerning personal liberty, fundamental rights, civil, criminal, service and commercial disputes, and other constitutional issues.

The problem of pendency cannot be gauged by just looking at the number of pending cases. Instead, one must look at the rate at which cases are being instituted and disposed of, as well as the number of cases that have been pending for a certain amount of time. The disparities in the caseload of the various High Courts require an empirical analysis, as the variation in their jurisdiction, strength, infrastructure, geographical location and other factors contributes to this issue. The growing number of cases might suggest that the number of cases disposed of is not on par with the number of cases being filed.

The current study therefore looks at the case pendency situation in terms of the number of cases pending in the quinquennial period and attempts to find whether there has been any change in the pendency scenario in that period. The study also intends to look at the percentage change in case pendency trends, as well as the High Courts which witnessed higher or lower trends than the others. The main question the current study hopes to address is whether the mechanisms currently in place are sufficient to handle the pendency of cases.

The problem of delay in the judicial system is not limited simply to the number of pending cases, but also extends to the cost that is incurred along with the delay. Apart from the costs associated with the litigation, the delay also affects the morale and confidence of the people in the system as well as their fundamental right to speedy justice as enshrined in Article 21 of the Constitution. Therefore, it is pertinent that the current study addresses the need for judicious reforms in order to expedite the justice delivery mechanism without affecting the quality of justice delivered.

Literature Review

The issue of judicial delay and case pendency in India has been the subject of discussion amongst many constitutional law scholars, researchers, committees and agencies. The existing literature highlights that the issue of perpetual pendency of cases is multifaceted and emerges on account of the rising volume of litigation coupled with an inadequate number of judicial and support personnel, complicated procedures, and a lack of infrastructure and case-management techniques.

The Law Commission of India has devoted several of its reports to the issue of arrears and delay in judicial proceedings and has recommended strengthening of the judiciary, better case administration and procedural reforms to achieve the goal of expeditious justice delivery. Besides emphasising the need for increasing the number of judges, the Commission has pointed out that without proper support systems, the problem of arrears, backlog and delay in case disposal would remain unaddressed.

In *Imtiyaz Ahmad v. State of Uttar Pradesh* (2012)[1], the Supreme Court highlighted the importance of examining the factors that contribute to judicial delay. In the same way, the Law Commission of India's 245th Report (2014)[2], titled *Arrears and Backlog: Creating Additional Judicial (Wo)manpower*, studied the relationship between manpower and the build-up of pending cases. These works lay a foundation for understanding the institutional dimensions of judicial pendency. Yet a separate five-year comparison of pendency trends across High Courts for the period 2021–2025 is needed to evaluate newer developments.

The National Judicial Data Grid (NJDG) is another crucial initiative that has thrown light on the state of judicial pendency by providing statistical information on the

number of instituted, disposed of and pending cases. Through comparative studies, we are now in a position to analyse the data on case pendency in the Indian judiciary and understand that the problem is not identical in every State or High Court.

The literature related to the interpretation of Article 21 of the Constitution provides us with various judgments and opinions regarding the right to speedy trial and justice. In several cases, the Courts have held that unjustified delay in criminal cases has detrimental effects on the accused as well as the victim. Several researchers have stressed that judicial time and speed must be considered a part of the fundamental rights.

Other literature related to vacancies and workload also contributes to our understanding of the problem. Articles related to vacancies and staff strength reveal that the shortage of judges and staff causes delays and pendency. At the same time, the literature also indicates that vacancies alone cannot be held responsible for the accumulation of arrears.

Studies on technological interventions and Alternate Dispute Resolution (ADR) initiatives taken up by the judiciary provide us with sufficient information regarding the measures being taken to address the issue of pendency. Virtual hearings, e-filing, digitalisation, case management software, automatic listing and online status tracking of cases are some of the technological interventions which reduce the workload and help in faster disposal of cases. Likewise, various ADR mechanisms, including Lok Adalats, mediation and arbitration, are effective in removing trivial cases from the judicial backlog, so that the Courts can focus on more serious matters.

Although there is an extensive literature on judicial delay, we still feel that a comparative study of quinquennial trends of pendency of cases in individual High Courts would help us understand the pattern and tendency of judicial delay. Most of the articles do not dwell on the issue of pendency in detail; instead, they comment on the issue in passing or discuss the problem in detail in one particular State or one particular area. We attempt a detailed empirical study of case-pendency trends in order to identify the patterns and causes of arrears and suggest appropriate remedial measures.

Research Gap

Despite extensive discussions in academic circles and institutions about judicial delay and case pendency in India, there is an absence of studies focusing on the comparison of pendency in the High Courts. The existing literature addresses either the general issues relating to judicial backlog or speediness of justice, vacancies, judicial reforms, or the very nature of speedy trial. But there is a lacuna in the research literature pertaining to pendency in High Courts on a yearly basis over a span of five years.

A comparison of the increase or decrease in case pendency of various High Courts was not done because just mentioning the number of pending cases does not suggest whether a court is performing better or worse over time.

There is a need to highlight the increase or decrease in the percentage of case pendency of various High Courts year over year to understand the changes. There is also scope to analyse and compare the trends of percentage increase or decrease of High Courts to identify whether some of them are showing a particular pattern.

Moreover, most of the literature either discusses judicial pendency from a theoretical and doctrinal perspective or suggests a reform agenda. There is a scarcity of empirical studies which utilise the publicly accessible data on judicial websites for analysing case pendency, institution and disposal trends. Judicial pendency analysis in India is in dire need of systematic evidence to support the reform agenda.

Additionally, the discussions on the need for fast-track courts mostly ignore the implications of case pendency trends for achieving the expeditious adjudication of justice enshrined in Article 21 of the Constitution of India. There is a need to understand the extent to which the five-year pendency trends can support the need for specific institutional, case-management and technology-enabled reforms, alternate dispute resolution mechanisms and other administrative measures. This would help in developing an evidence-based law-reform agenda to reduce backlog without compromising on justice, rights, accountability, transparency and judicial independence.

Therefore, the current research study attempts to bridge the research gap by focusing on the five-year caseload trends of the pendency of cases in the High Courts of India. It compares the changes in case pendency year over year, analyses the trends, discusses their implications for the judiciary and develops a set of recommendations to address the concerns.

The study's primary research gap pertains to the lacuna in the analysis of five-year pendency trends of High Courts in India and their implications for the fast-track adjudication system and speedy trial.

Research Methodology

The present study, entitled "A Quinquennial Empirical Analysis of Case Pendency in Indian High Courts and the Imperative for Expedited Adjudication", is descriptive, empirical, quantitative and comparative research. It

focuses on analysing the five-year case pendency, including its implications for quick disposal, of various Indian High Courts.

The study makes heavy use of secondary data, including the National Judicial Data Grid (NJDG), the Department of Justice, Government of India, annual reports, High Court reports, Law Commission reports and related literature. It uses classification, presentation of data in tabular form, the calculation of percentages, comparisons, graphs, etc.

It calculates the increase or decrease in percentage pendency on a year-to-year basis by taking the ratio of the number of cases pending in the current year to the previous year, and multiplying it by 100 to represent the change. Moreover, it discusses relevant statutes, constitutional provisions and case laws related to the pendency of cases, the right to speedy justice, access to justice, justice under Article 21 and so on, to analyse the nexus between empirical results and theoretical perspectives. Thus, the study primarily focuses on examining the extent and nature of case pendency in the Indian judicial system and its implications for speedy trials.

Purpose of the study

The main objective of the present study is to analyse the trend and pattern of case pendency of different Indian High Courts over a period of five years and understand the implications. It also discusses various statutory, judicial and policy-level suggestions for fast-tracking of cases and expeditious adjudication.

The study tries to:

1. Explore the changes in the number of pending cases of various Indian High Courts from year to year;
2. Identify the High Courts where pendency is rising, falling or fluctuating;
3. Calculate the percentage change in pendency for these High Courts from year to year;
4. Discuss the impact of caseload on judicial efficacy and the Indian judicial system;
5. Discuss the concept of speedy trial and its implications for the Indian judicial system;
6. Analyse the relationship between case pendency and the right to speedy justice; and
7. Suggest pragmatic solutions for quick judicial delivery without affecting the rights of the accused and the fairness of the trial.

Research questions

1. What has the year-by-year trend of case pendency been in High Courts over the selected five-year period?
2. What percentage increase or decrease in case pendency has taken place across the High Courts during the study period?
3. Which High Courts have kept seeing increases or decreases in case pendency, and what comparative patterns can be spotted?
4. How much does persistent case pendency impact the goal of quick and easy justice under Article 21?
5. What institutional, procedural, technological and administrative factors keep case pendency going?
6. What steps can be taken to speed up adjudication while keeping fairness, due process and high quality in decisions?

Research objectives

The study has the following objectives:

1. To examine how case pendency changes over five years in High Courts.
2. To analyse how the number of pending cases shifts each year, and by what percentage, during the chosen period.
3. To compare the pendency position of High Courts and spot big differences.
4. To find patterns where pendency goes up, goes down or swings around in the High Courts.
5. To look at the institutional and procedural reasons that cause judicial backlog, such as vacant judge posts, more litigation, limits in infrastructure, slow procedures and weak case management.
6. To explore the impact of long pendency, especially how it touches Article 21 and the rule that justice must be fast.
7. To assess how technology, e-Courts, alternative dispute resolution and good case-flow management can cut backlog.
8. To recommend proven steps that speed up decisions yet keep fairness, independence and due process.

Hypotheses

Null hypothesis (H_0): There is no change in the number of pending cases in Indian High Courts during the chosen five-year period. The changes that do happen do not show a trend that needs major changes in the system to speed up court decisions.

Alternative hypothesis (H_1): There is a significant change in the number of pending cases in Indian High Courts during the chosen five-year period. Overall, the

number is going up, which means we need changes in how the courts work, including procedures, technology, administration and structure, to speed up the justice process.

Additional hypotheses:

- **H_2:** High Courts that keep seeing pending cases over time have a harder time giving people justice quickly. This makes it harder to follow the Constitution's promise of justice for everyone.
- **H_3:** Making the judiciary stronger, improving how cases are managed, cutting down on delays, using digital tools and encouraging alternative ways to resolve disputes can all help lower the number of pending cases and speed up court decisions.
- **H_4:** Speeding up court decisions can reduce the backlog of cases without hurting fairness or due process. This only works if the reforms are backed by enough judges, staff, technology and clear rules to protect everyone's rights.

Scope, Significance and Relevance of the Study

The study "A Quinquennial Empirical Analysis of Case Pendency in High Courts and the Imperative for Expedited Adjudication" looks at the actual data on case pendency over five consecutive years. It examines how many cases are pending each year, the rise or fall in numbers, differences between High Courts and how delays affect the right to justice.

Case pendency is not merely a matter of court records. It touches the core of the rule of law, the rights of people who go to court and how much trust the public places in the judiciary. The National Judicial Data Grid (NJDG) collects data on case disposal and backlog across High Courts and lower courts. This makes it possible to measure delays with numbers, not just guesswork.

The study matters because of Article 21 of the Constitution. The Supreme Court has said this article includes the right to a speedy trial. So this study does more than just count pending cases. It checks whether the current systems in courts and administration are working well enough to deliver justice on time.

International Perspective

Judicial delay is a problem that affects countries around the world. Each legal system has tried to find its own way to deal with it. In the United States, the Sixth Amendment clearly says that people accused of crimes have the right to a speedy and public trial. Because of this, American courts have created rules about what counts as too much delay, how to manage cases and how to protect people from staying in jail for too long before their trial.

In the United Kingdom, courts have worked to make the legal process faster and more efficient. They use tools like managing cases actively, having digital courts, using electronic filing, finding out early what the arguments are about and using other ways to settle disputes without going to court. The goal is to make sure that legal cases move forward in a timely way.

In Australia, judges have taken an active role in managing cases. Courts control how cases are handled, when they take place, what evidence is used and when they can be put off. These examples from other countries show that reducing delays in courts needs a mix of things, like having enough judges, following the rules carefully, using technology, managing cases well and using other ways to settle disputes. It is not about one single change.

For India, copying what other countries do is not enough. The solutions must fit India's Constitution, its large population, the huge number of cases, the different economic and social backgrounds of people and the abilities of its courts.

Comparative Analysis

A comparison of High Courts is essential because the number and type of pending cases vary across courts.

The study looks at High Courts using the following points:

1. Year-wise pending cases;
2. Annual decrease in pendency;
3. Percentage change over the five-year period;
4. Cumulative quinquennial change;
5. Increasing, decreasing or fluctuating trends;
6. Relative institutional burden wherever comparable data is available.

This analysis helps us see the difference between a High Court that has a large number of cases at once and a High Court whose backlog is growing fast. Research using data has also found big differences in caseloads among High Courts and has noted that pendency trends and judicial workload vary a lot between jurisdictions. Therefore, comparing High Courts gives a more meaningful picture than simply ranking them by the absolute number of pending cases. It can show which High Courts need more judges, better infrastructure, improved case-flow management, specialised benches or targeted disposal strategies.

Indian Judicial Context

The Indian judiciary has played a role in shaping the constitutional principle of speedy justice. The Supreme Court's decisions show that unreasonable delay can violate Article 21. In *Hussainara Khatoon v. Home Secretary, State of Bihar* (1979), the Supreme Court said

that a speedy trial is part of Article 21 and warned that keeping prisoners for too long can violate their fundamental rights. The Court also pointed out the need for more courts, judges, staff and facilities to tackle delay. In *Abdul Rehman Antulay v. R.S. Nayak* (1992)[3], the Supreme Court examined the right to a speedy trial in detail and explained that the requirement must be understood in the context of each case. The Court did not set a time limit that applies to every criminal proceeding. In *P. Ramachandra Rao v. State of Karnataka* (2002)[4], the Supreme Court clarified that courts should not apply fixed time limits mechanically in every case. Speed must be balanced against the need for a fair trial and the specific circumstances of each proceeding. Later Supreme Court decisions continued to confirm that a speedy trial covers the criminal-justice process, including investigation, trial, appeal, revision and retrial. Thus, the role of the judiciary is not to dispose of cases quickly but to achieve reasonably fast adjudication that is fair, respects natural justice and follows due process.

The Indian judicial context shows that pendency is both a constitutional and an institutional problem. This current empirical study is especially relevant because official judicial data still shows a large High Court caseload. For example, a Government of India parliamentary answer based on data reported 63,63,406 cases pending before High Courts as of 8 December 2025.

Hussainara Khatoon v. Home Secretary, State of Bihar (1979)[5]

This important case set the foundation for the right to a speedy trial under Article 21 of the Constitution. The Court said that keeping someone in jail for a long time without a trial goes against their basic rights. It also stressed that the justice system needs judges, courts and support to handle cases properly.

Relevance: It shows that delays in court are not a small problem. They are a serious constitutional issue.

Abdul Rehman Antulay v. R.S. Nayak (1992)[6]

A group of five judges looked closely at what “speedy trial” means. They gave rules to decide if a delay is too long and breaks Article 21. They said judges must look at how long the delay is, why it happened, what the accused did during that time and how much harm the delay caused. They warned against using time limits that do not consider the facts of each case.

Relevance: This case teaches us that justice must be fast but fair. We cannot rush things just to meet a deadline.

P. Ramachandra Rao v. State of Karnataka (2002)[7]

The Supreme Court said that we cannot set a fixed time limit for every case and automatically stop the trial if it goes beyond that. Justice must be decided based on what is in each case, not by following a one-size-fits-all rule. The right to a speedy trial must work with the need to do things properly and fairly.

Relevance: This reminds us that fast justice does not mean justice.

Vakil Prasad Singh v. State of Bihar (2009)[8]

The Court made it clear that the right to a speedy trial is not just about what happens in court. It also includes the investigation stage before the trial begins. People have a right to a fair investigation too.

Relevance: This means we must fix delays not only during trial but also earlier, in police work and evidence collection.

Contemporary judicial context

Today courts have tools to track how many cases are pending. The National Judicial Data Grid gives information about each court: how many cases are waiting, how many are finished and whether they are civil or criminal.

This study uses those numbers and connects them with past court decisions. Numbers show where the backlog is growing. Case law explains why long delays matter under the Constitution. Together, this helps plan fixes, like hiring more judges, building better court facilities, using digital systems, managing cases better, reducing unnecessary delays, promoting alternative dispute resolution and giving priority to old cases.

Overall relevance

The central proposition emerging from the judicial context is that speed and justice must operate together. The objective of expedited adjudication should not be to maximise disposal figures at the expense of hearing and reasoned decision-making. Rather, the objective should be to create a judicial system in which cases are resolved within a reasonable time, through a fair procedure, with adequate judicial resources and effective institutional management. This principle provides the jurisprudential and empirical foundation for the present quinquennial analysis.

Discussion

Allahabad High Court

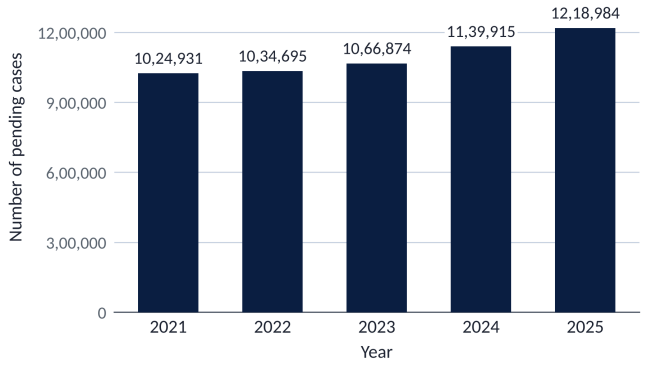


Figure 1. Year-wise pending cases in the Allahabad High Court, 2021–2025

The year-by-year look at case pendency at the Allahabad High Court shows a rise between 2021 and 2025. In 2021 there were 10,24,931 pending cases. By 2025 that number went up to 12,18,984. That is an increase of about 18.93%.

The growth was slow at first: 0.95% in 2022. Then it picked up. In 2023 the increase was 3.11%. In 2024 it jumped to 6.84%. In 2025 it reached 6.94%.

These numbers show the court system is under pressure. There is a need for better case management, more judges and court staff, better technology, faster investigations and changes in how cases are handled. All of this is needed to cut down on the number of pending cases and get cases resolved quickly.

Bombay High Court

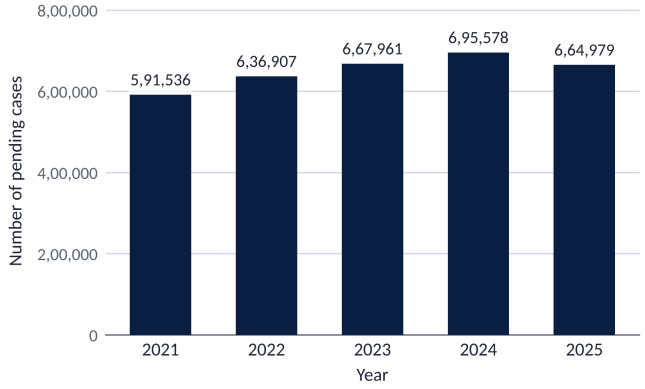


Figure 2. Year-wise pending cases in the Bombay High Court, 2021–2025

The year-by-year study of case pendency in the Bombay High Court shows a rise from 2021 to 2024 and then a drop in 2025. Pendency grew from 5,91,536 cases in 2021 to 6,95,578 cases in 2024, with the largest yearly jump of 7.67% seen in 2022. The pace of growth slowed in 2023 and 2024. The year 2025 saw a 4.40% fall in pendency. After this recent cut, pendency in 2025 stayed 12.41% above the 2021 figure. This drop might point to court disposal and case-management efforts, yet the still-high volume of pendency shows that ongoing procedural, technological and institutional changes are needed to bring faster justice.

Calcutta High Court

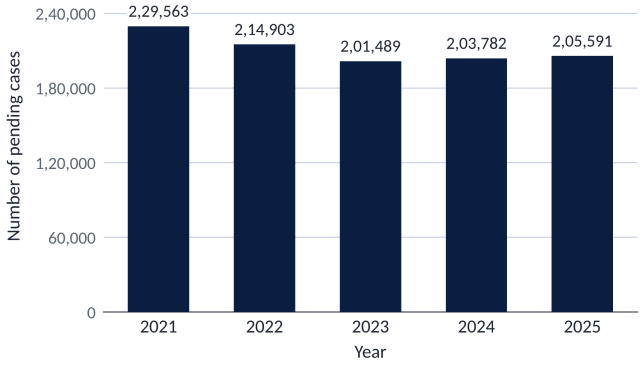


Figure 3. Year-wise pending cases in the Calcutta High Court, 2021–2025

The year-wise analysis of pendency in the Calcutta High Court shows a decline from 2021 to 2025. Pendency dropped by 6.39 percent in 2022 and by another 6.24 percent in 2023, cutting the number of cases from 2,29,563 in 2021 to 2,01,489 in 2023. In 2024 and 2025 the trend turned a little upward, with pendency rising by 1.14 percent in 2024 and by 0.89 percent in 2025. So the total pending cases in 2025 are still about 10.44 percent fewer than in 2021. This overall fall shows that the Calcutta High Court is disposing of cases well and managing its docket effectively. The Calcutta High Court is doing a good job, yet the recent rise in pendency means that the Calcutta High Court must keep working hard to stop new cases from piling up and to keep justice fast.

Gauhati High Court

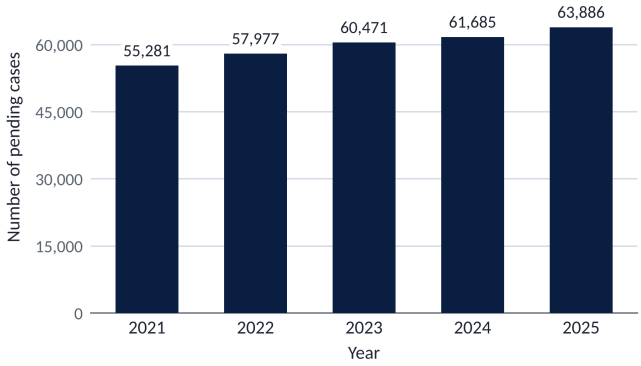


Figure 4. Year-wise pending cases in the Gauhati High Court, 2021–2025

The year-wise analysis of pendency in the Gauhati High Court shows a rise from 2021 to 2025. Pendency grew from 55,281 cases in 2021 to 63,886 cases in 2025, a jump

of 15.57 percent. The biggest yearly jump happened in 2022 at 4.88 percent, while the smallest rise was in 2024 at 2.01 percent. Even though the growth slowed in 2024, pendency went up again by 3.57 percent in 2025. This steady increase in pendency keeps the court busy and shows that we need better case management, more judges, technology and changes to procedures so that cases can finish faster.

High Court for the State of Telangana

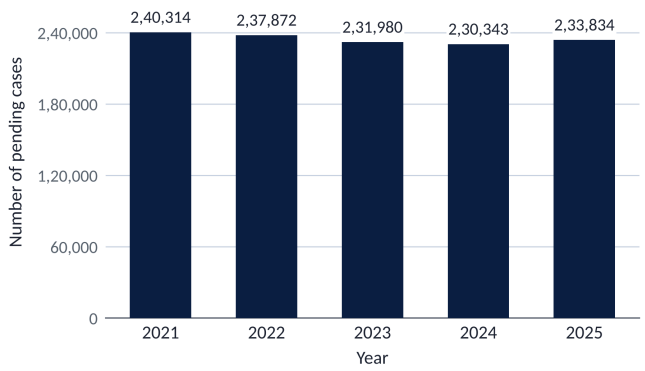


Figure 5. Year-wise pending cases in the High Court for the State of Telangana, 2021–2025

The data shows that the number of cases waiting to be handled went down for the first four years. In each of those years the number dropped: by 1.02%, then 2.48% and then 0.71%. The number of cases was the smallest in the fourth year. It was 2,30,343 cases. In the fifth year the number went up by 1.52%. It went to 2,33,834 cases. Even though there was an increase in the fifth year, the total over five years still shows that the number of cases waiting to be handled went down by 2.70%.

High Court of Andhra Pradesh

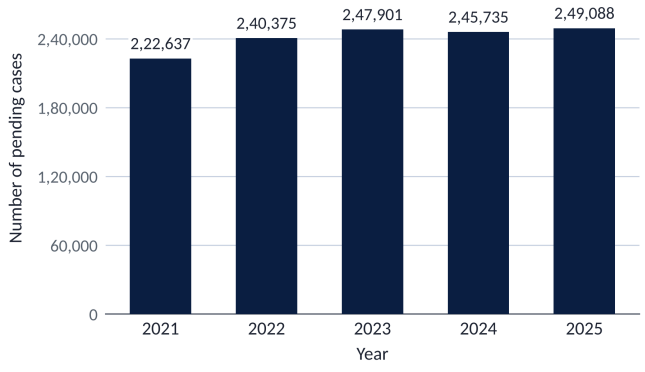


Figure 6. Year-wise pending cases in the High Court of Andhra Pradesh, 2021–2025

The data shows that case pendency in the High Court of Andhra Pradesh grew a lot during the first three years.

The biggest jump happened from Year 1 to Year 2, with a rise of 7.97%. This means that pendency grew a lot.

In Year 3 pendency rose again, this time by 3.13%, bringing the total to 2,47,901 cases.

Year 4 saw an improvement; pendency dropped by 0.87%, falling to 2,45,735 cases.

That drop did not last. In Year 5 pendency climbed again by 1.36%, reaching the highest level of the five years, 2,49,088 cases.

High Court of Delhi

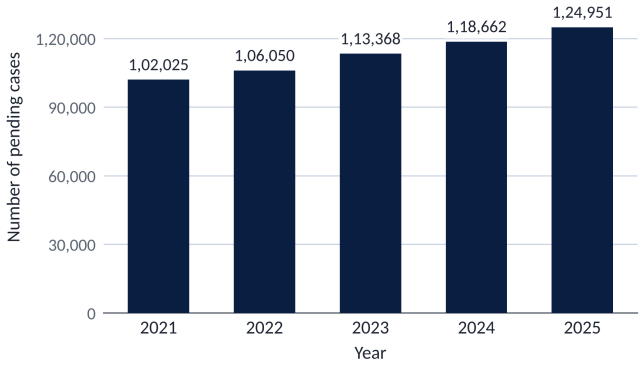


Figure 7. Year-wise pending cases in the High Court of Delhi, 2021–2025

This data shows that pendency keeps growing each year of the study period. Pendency grew by 3.94 percent in the second year and then grew even more, by 6.90 percent, in the third year. The growth slowed to 4.67 percent in the fourth year, but pendency grew again by 5.30 percent in the fifth year.

It is especially important that no year shows a decline. The pending caseload rose from 1.02 lakh to about 1.25 lakh cases, an increase of almost 22.5 percent. This shows that the rate at which cases accumulate is higher than the rate at which cases are being disposed of during the period.

High Court of Gujarat

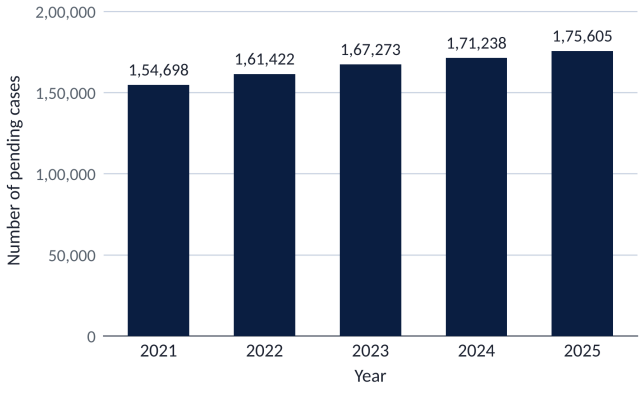


Figure 8. Year-wise pending cases in the High Court of Gujarat, 2021–2025

The data shows a rise in pendency. The biggest jump happened from Year 1 to Year 2, when pendency grew by 4.35%. After that the growth rate fell to 3.63% in Year 3

and 2.37% in Year 4. In Year 5 the growth rate went up a bit to 2.55%.

There was no year in the five-year span where pendency fell. Even though the growth rate slowed after Year 2, the total number of pending cases kept growing each year.

The High Court of Gujarat saw a rise of 20,907 pending cases, a 13.52% increase over five years. This steady rise means that case disposal has not kept up with existing litigation.

High Court of Himachal Pradesh

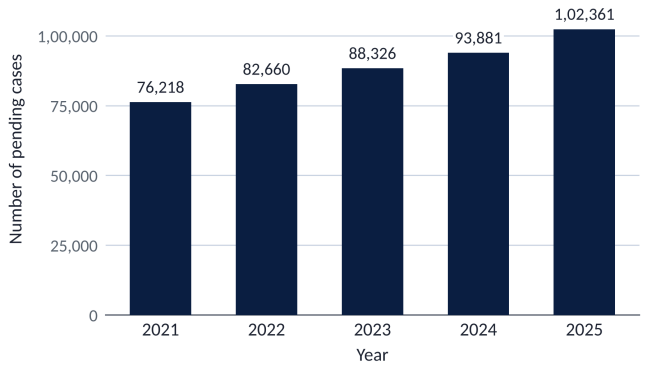


Figure 9. Year-wise pending cases in the High Court of Himachal Pradesh, 2021–2025

The High Court of Himachal Pradesh shows a rise in case pendency over a five-year span. Pending cases grew from 76,218 in the first year to 1,02,361 in the fifth year, an absolute increase of 26,143 cases, which equals a total rise of 34.30 percent. Year by year, pendency grew by 8.45 percent, 6.85 percent, 6.29 percent and 9.03 percent. The largest yearly rise happened in the fifth year at 9.03 percent. No year shows a drop, which means cases keep piling up. This pattern underlines the need for greater judicial capacity, better case-management systems, faster disposal, more digital tools and alternative dispute-resolution methods to speed up judgments.

High Court of Jammu and Kashmir

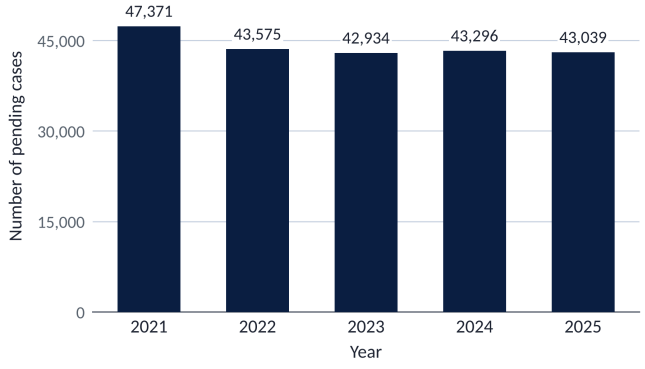


Figure 10. Year-wise pending cases in the High Court of Jammu and Kashmir, 2021–2025

The data shows a reduction of 9.14% in case pendency over the five-year period. The biggest drop happened between Year 1 and Year 2, when pendency decreased by 8.01%. In Year 3 there was a drop of 1.47%. In Year 4 pendency went up slightly by 0.84%, which means there was a short-term reversal. Then in Year 5 pendency fell again, this time by 0.59%.

Overall, the High Court of Jammu and Kashmir shows a trend of case pendency going down over time. Even though there were ups and downs each year, the long-term direction is clearly downward.

High Court of Chhattisgarh

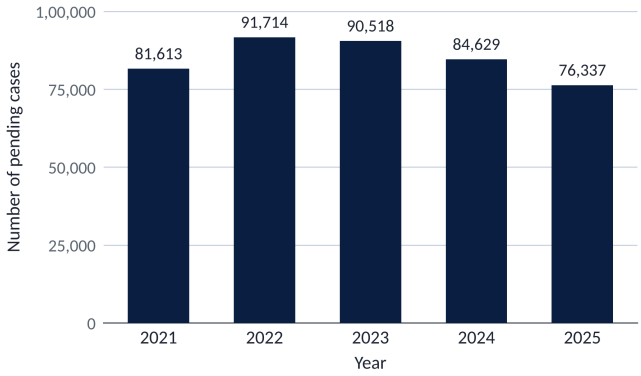


Figure 11. Year-wise pending cases in the High Court of Chhattisgarh, 2021–2025

The High Court of Chhattisgarh saw a drop of 5,276 pending cases, which is a 6.46% decrease over five years. In the second year the number of pending cases went up sharply by 12.38%. After that the trend reversed. The next three years showed declines of 1.30%, 6.50% and 9.80% respectively. This shows that the High Court of Chhattisgarh is making progress in clearing cases. The data suggests that the court is moving in the right direction. It also highlights the importance of keeping up with good case management. Timely disposal of cases is key. Digital judicial tools can help speed things up. Alternative dispute-resolution methods should also be used more. These steps together can help achieve justice.

High Court of Jharkhand

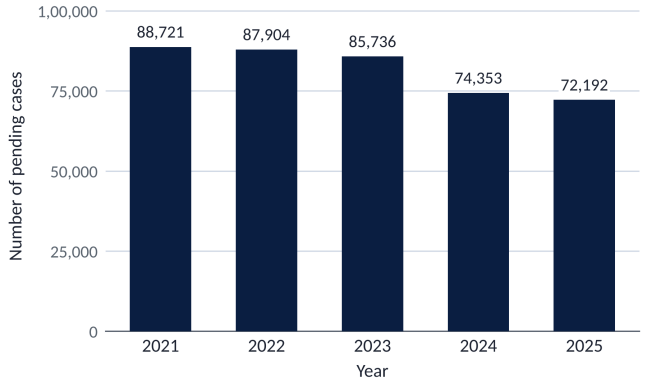


Figure 12. Year-wise pending cases in the High Court of Jharkhand, 2021–2025

The data shows that the High Court of Jharkhand had a drop in case pendency over five years. At first there was a drop of 0.92% in the second year. Then in the third year there was a bigger drop of 2.47%. In the fourth year the drop was much larger, with pending cases going down by 13.27%. This was the largest drop in one year during the whole time period. In the fifth year there was another drop, this time of 2.91%.

In total, the number of pending cases went down by 16,529. That is an 18.63% decrease over five years. While other High Courts had rising numbers, the High Court of Jharkhand had a drop in the number of cases it had to deal with.

High Court of Karnataka

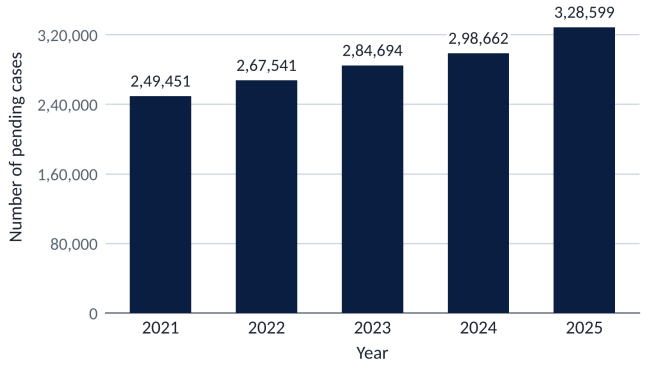


Figure 13. Year-wise pending cases in the High Court of Karnataka, 2021–2025

The data shows a rise in the number of pending cases before the High Court of Karnataka. In the second year the pendency went up by 7.25%. The increase continued in the third year with a rise of 6.41%. Although the growth slowed a bit in the fourth year to 4.91%, the final year saw a sharp jump, with pendency increasing by 10.02%.

What stands out most in the data is that there was no year when pendency decreased. The number of pending cases grew from around 2.49 lakh cases to 3.29 lakh cases. This means the total increase is 31.73%. This big jump shows that the court has not been able to clear cases fast enough. The number of cases has kept building up faster than they can be resolved.

High Court of Kerala

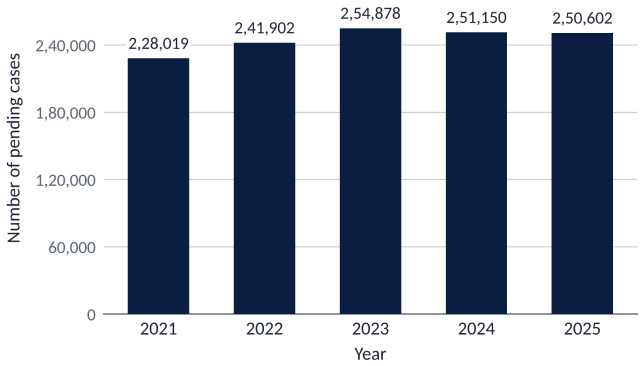


Figure 14. Year-wise pending cases in the High Court of Kerala, 2021–2025

The data shows that case pendency in the High Court of Kerala went up a lot at first. Pendency went up by 6.09% in the second year and then by another 5.36% in the third year. This made the number of cases reach a high of 2,54,878. This means that a lot of cases built up during the first part of the study period.

The pattern changed in the fourth year, when pendency went down by 1.46%. In the fifth year it went down a little more, by 0.22%. These drops show that there was some progress in managing the backlog. The drops were not enough to cancel out the big increases that happened in the first three years. So over the five years there was still a 9.90% rise in pending cases. The data shows a mix of changes, but more cases are pending. The recent small improvements need support from the institutions to keep going.

High Court of Madhya Pradesh

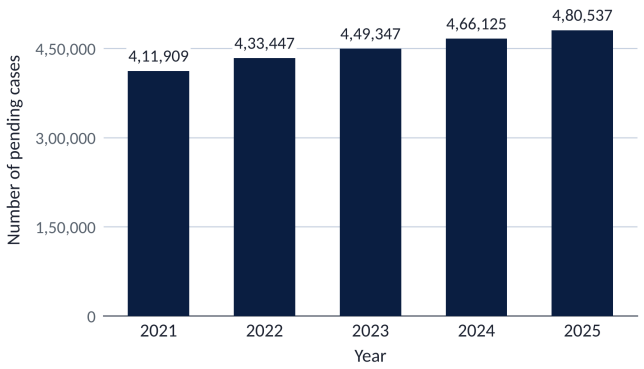


Figure 15. Year-wise pending cases in the High Court of Madhya Pradesh, 2021–2025

The data shows that the number of cases waiting to be heard at the High Court of Madhya Pradesh kept going up during the five-year period. In the second year the number of pending cases went up by 5.23%. Then in the following years it went up by 3.67%, 3.73% and 3.09% respectively. Even though the amount of increase got smaller after the second year, the total number of cases that were still pending went up every single year.

There was no year where the number of pending cases went down. The number of cases that were not yet decided went from about 4.12 lakh to 4.81 lakh. This means there was an increase of 68,628 cases, or 16.66%. This shows that the rate at which cases were being resolved was not enough to stop the growing number of cases that were still waiting.

High Court of Manipur

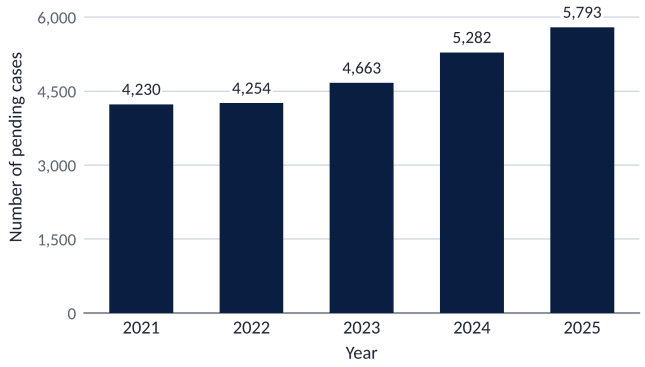


Figure 16. Year-wise pending cases in the High Court of Manipur, 2021–2025

The data shows that there was a rise in the number of cases that were not resolved in the High Court of Manipur over five years. In the second year the rise was small, at 0.57%, but then it grew a lot after that. In the third year the number of cases that were still pending went up by 9.62%. In the fourth year the highest increase happened, at 13.27%. In the fifth year the number of pending cases went up again by 9.67%.

No year had a drop in the number of pending cases. The total number of cases that were still pending went up by 1,563 cases, which is a big increase of 36.95% overall. The fast growth shows that more and more cases are piling up. This creates worries about whether the court system can handle and settle cases fast enough to avoid a big backlog.

High Court of Meghalaya

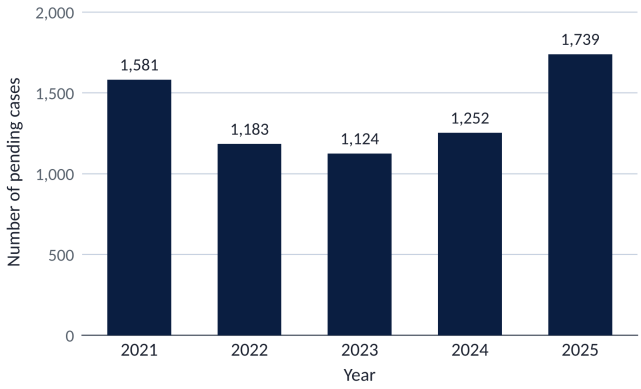


Figure 17. Year-wise pending cases in the High Court of Meghalaya, 2021–2025

The High Court of Meghalaya shows a variable pattern of case pendency over a five-year period. Pendency fell sharply by 25.17 percent, dropping from 1,581 cases to 1,183 cases in the second year. A further drop of 4.99 percent happened in the third year, when pendency hit its lowest point at 1,124 cases.

The falling trend later turned around. Pendency rose by 11.39 percent in the fourth year and jumped sharply by 38.90 percent in the fifth year, reaching 1,739 cases, the highest number seen in the period. The rise in the final year is especially important because it more than made up for the earlier drops.

In total, pendency grew by 158 cases, which is a net increase of 9.99 percent over five years. The data shows that although big reductions were made at first, they did not last.

High Court of Punjab and Haryana

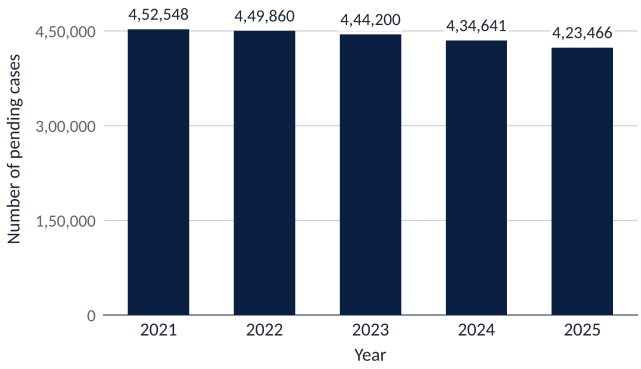


Figure 18. Year-wise pending cases in the High Court of Punjab and Haryana, 2021–2025

The High Court of Punjab and Haryana shows an uneven pattern of case backlog over five years. Backlog dropped each year: in the second year it fell by 0.59%, in the third by 1.26%, in the fourth by 2.15% and in the fifth by 2.57%. The fact that backlog never increased is important. The number of pending cases went from 4,52,548 to

4,23,466, a drop of 29,082 cases or 6.43%. The larger yearly drops in later years show that the court is getting better at handling and closing cases.

High Court of Rajasthan

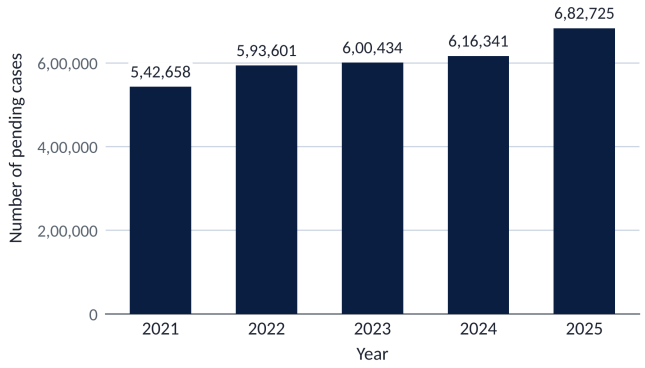


Figure 19. Year-wise pending cases in the High Court of Rajasthan, 2021–2025

The data shows that the number of cases kept growing before the High Court of Rajasthan. The biggest jump happened between Year 1 and Year 2, when the cases increased by 9.39%. Then there was a rise of 1.15% in Year 3. The number of cases went up by 2.65% in Year 4. The last year had a rise of 10.77%, making the total number of cases 6,82,725. That was the highest number during the whole study period.

Importantly, there was no year where the number of cases went down. The total number of cases went up by 1,40,067, which is 25.82%. That is an increase in the number of cases that are still waiting to be handled. The big rise in the final year stands out and shows that getting rid of the old cases is still a big problem for the court system.

High Court of Sikkim

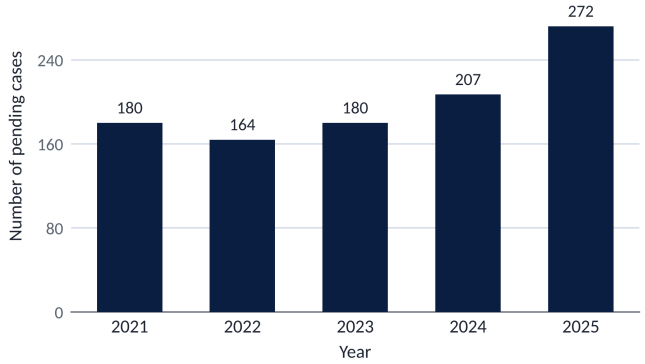


Figure 20. Year-wise pending cases in the High Court of Sikkim, 2021–2025

The data shows a changing pattern of case pendency in the High Court of Sikkim. In the second year pendency fell by 8.89%, dropping from 180 to 164 cases. That drop was brief, because pendency rose back to 180 cases in the third year, a 9.76% increase. After that the upward trend grew stronger. Pendency went up by 15.00% in the fourth year and by a large 31.40% in the fifth year, reaching 272 cases, the highest level seen in the study period. Even though the total number of cases is not very large, the 51.11% overall rise is big in percentage terms. The data shows that the High Court had a large proportional growth in its pending caseload.

High Court of Tripura

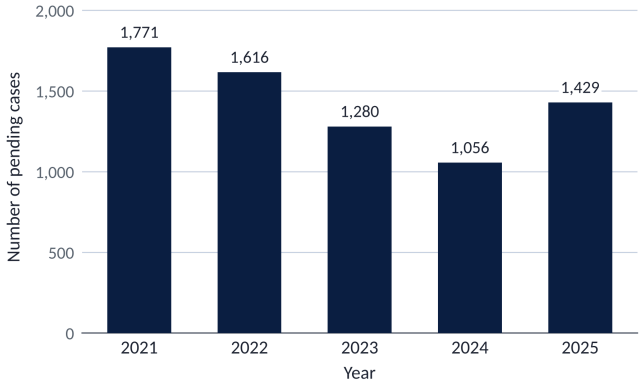


Figure 21. Year-wise pending cases in the High Court of Tripura, 2021–2025

The High Court of Tripura demonstrates a strong declining trend during the first four years of the study period. Pendency decreased by 8.75% in the second year, followed by a significant reduction of 20.79% in the third year and a further 17.50% decline in the fourth year. Consequently, pending cases reached their lowest level of 1,056 cases in the fourth year.

However, the fifth year witnessed a substantial reversal, with pendency increasing by 35.32%, rising to 1,429 cases. Despite this sharp final-year increase, the overall quinquennial position remains favourable, with pendency declining by 342 cases, equivalent to 19.31%.

High Court of Uttarakhand

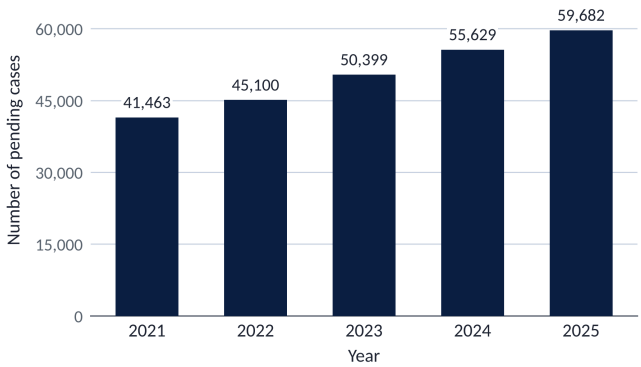


Figure 22. Year-wise pending cases in the High Court of Uttarakhand, 2021–2025

The data shows a year-to-year rise in case pendency in the High Court of Uttarakhand. Pendency grew by 8.77 percent in the second year, then jumped to 11.75 percent in the third year. The upward trend kept going, with a 10.38 percent rise in the fourth year and a 7.28 percent rise in the fifth year. Although the yearly growth rate fell in the later years, the total number of pending cases still went up. Importantly, no year saw a drop in pendency. Across the five-year stretch, pending cases rose by 18,219 cases, a big 43.94 percent overall increase.

Madras High Court

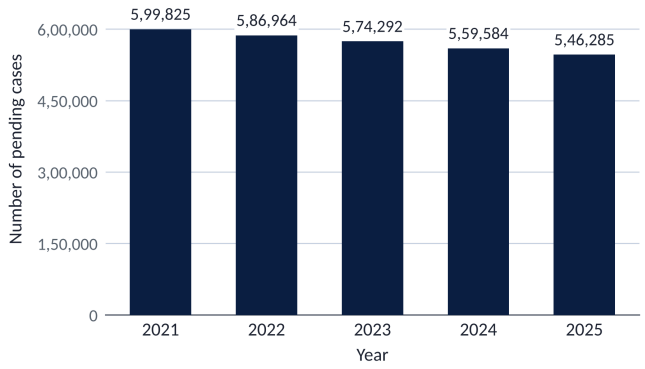


Figure 23. Year-wise pending cases in the Madras High Court, 2021–2025

The data shows a year-on-year drop in case pendency before the Madras High Court. Pendency fell by 2.14% in 2022, then fell again by 2.16% in 2023. The fall grew to 2.56% in 2024 before easing to 2.38% in 2025. Importantly, each year saw a drop, with no year showing a rise. Over the five-year span, pending cases were reduced by 53,540 cases, a drop of 8.93%. This steady decline shows an effort to keep the judicial backlog under control. The fall is especially important because the Madras High Court started this period with nearly six lakh pending cases.

Orissa High Court

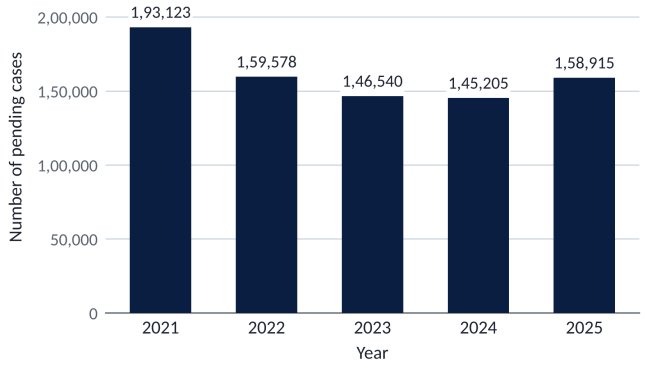


Figure 24. Year-wise pending cases in the Orissa High Court, 2021–2025

The data shows a drop in pendency during the first four years. The biggest drop in pendency happened in 2022, when pendency fell by 17.37%, from 1,93,123 to 1,59,578. Another drop of 8.17% in pendency happened in 2023, followed by a drop of 0.91% in 2024. Consequently, pendency reached a low point of 1,45,205 cases in 2024.

The good trend turned back in 2025, when pendency rose by 9.44%, reaching 1,58,915 cases. Even though pendency rose in 2025, the overall five-year picture stayed good because pendency fell by 34,208 cases, which is 17.71%. The Orissa High Court shows progress in cutting accumulated cases, especially in 2022 and 2023. The increase in 2025 shows that the drop in pendency was not sustained.

Patna High Court

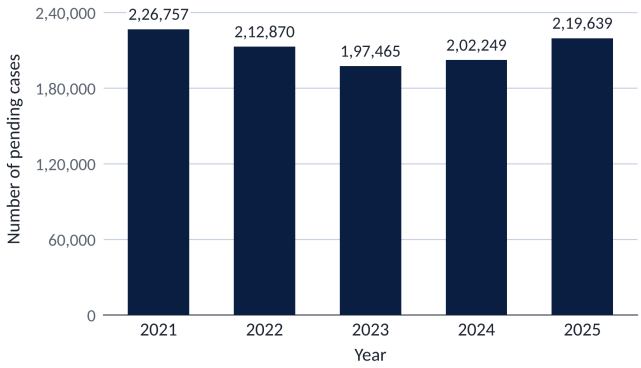


Figure 25. Year-wise pending cases in the Patna High Court, 2021–2025

The case backlog data for the Patna High Court shows a mixed pattern. In the first two years there was a big drop in the number of cases waiting to be decided. The backlog went down by 6.13% in 2022, then by another 7.24% in 2023, when it reached the smallest number of cases recorded in the study period, 1,97,465.

This downward trend stopped in 2024, when the number of cases went up by 2.42%. Then in 2025 there was a jump of 8.60%. At the end of the period studied, the number of cases was 2,19,639. Even though there was an increase in the last two years, the overall situation over five years still shows a 3.14% decrease compared to 2021.

All High Courts

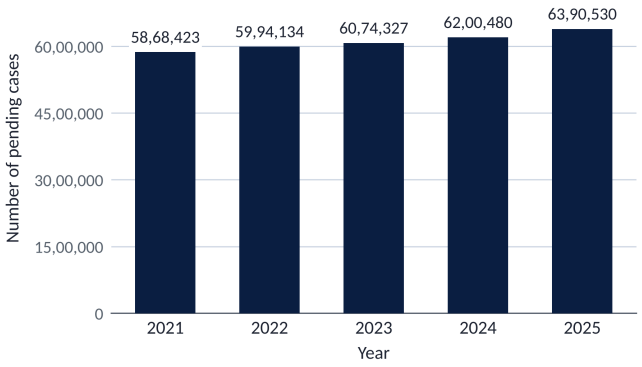


Figure 26. Total pending cases in all 25 High Courts, 2021–2025

The aggregate data shows a steady rise in pendency across all Indian High Courts. The total number of pending cases grew each year, with no year showing a decline over the five-year period. Pendency grew by 2.14% in 2022, then by 1.34% in 2023. Pendency grew again, by 2.08%, in 2024 and by 3.07% in 2025, the highest yearly increase in the study.

Overall, pendency rose by 5,22,107 cases, which is an 8.90% increase over five years. The sharp rise in 2025 is especially important because it shows that the total judicial workload keeps growing despite many reforms and new technology.

Results

The five-year data above gives a factual basis for checking the state of pendency in Indian High Courts and its effect on quick, fair and effective trials. Pendency grew from 58,68,423 cases in 2021 to 63,90,530 cases in 2025. So during the study period the total backlog grew by 5,22,107 cases, about an 8.90% rise.

The aggregate figures show that pendency grew each year. Even though the growth rate slowed to 1.34% in 2023, pendency then picked up again, to 2.08% in 2024 and 3.07% in 2025. This shows that the problem of growing litigation stays strong.

High Courts with the highest pendency

The data reveals considerable variation among High Courts. In 2025, the largest pending caseloads were recorded by:

1. Allahabad High Court – 12,18,984
2. Rajasthan High Court – 6,82,725
3. Bombay High Court – 6,64,979
4. Madras High Court – 5,46,285
5. Madhya Pradesh High Court – 4,80,537
6. Punjab and Haryana High Court – 4,23,466
7. Karnataka High Court – 3,28,599

These figures show that the burden of pendency is not spread evenly across the High Court system. The Allahabad High Court alone jumped from 10,24,931 cases in 2021 to 12,18,984 in 2025, a rise of 18.93 percent. The Rajasthan High Court increased by about 25.82 percent and the Karnataka High Court increased by 31.73 percent. The burden of pendency is clear in each of these courts.

High Courts showing reduction in pendency

The empirical data also demonstrates that some High Courts succeeded in reducing their accumulated caseload.

Particularly notable reductions between 2021 and 2025 include:

- Tripura: −19.31%
- Jharkhand: −18.63%
- Orissa: −17.71%
- Jammu & Kashmir: −9.14%
- Madras: −8.93%
- Chhattisgarh: −6.46%
- Punjab and Haryana: −6.43%
- Telangana: approximately −2.70%

These results are important because they demonstrate that reduction of pendency is achievable through effective institutional and case-management practices. However, the aggregate national position remains adverse because increases in several high-volume High Courts outweigh reductions elsewhere.

Implications for speedy and fair trial

The empirical results have implications for the constitutional principle of speedy justice under Article 21. A backlog that keeps growing can lead to litigation, create more uncertainty for litigants, delay remedies for victims and add more burdens on accused persons in criminal proceedings.

At the same time, speedy adjudication cannot be the same as disposing of cases at any cost. An effective judicial system must achieve both speed and fairness, due process and quality of adjudication. Speedy adjudication is essential. Fairness must not be sacrificed.

The goal should be expedited adjudication, not just quick disposal of cases. Excessive pressure to dispose of cases quickly can threaten fairness if it limits hearing, legal representation, evidence examination or thoughtful decision-making.

Final research result

The five-year empirical analysis of case pendency in High Courts shows that the judicial backlog has kept growing from 2021 to 2025. Total pendency rose from 58,68,423

cases to 63,90,530 cases, an increase of 5,22,107 cases and a growth of 8.90%. While several High Courts, especially Tripura, Jharkhand, Orissa, Jammu & Kashmir and Madras, cut back on cases, these gains were not enough to balance the rises in high-volume areas such as Allahabad, Rajasthan, Bombay, Karnataka and Madhya Pradesh. The data prove that a speedy and fair trial needs more than disposal. It needs institutional reform that grows judicial capacity, improves case management, uses technology, boosts procedural speed and protects due-process rights. The evidence clearly shows that we must take steps to achieve expedited adjudication without losing fairness or the quality of justice.

Major Findings of the Study

The quinquennial empirical analysis of case pendency in the 25 High Courts of India for the period 2021–2025 reveals significant variations in the judicial workload and disposal trends. The findings are particularly relevant to the objective of ensuring speedy, fair and effective adjudication.

Overall increase in judicial pendency

The most significant finding is that aggregate pendency increased continuously from 58,68,423 cases in 2021 to 63,90,530 cases in 2025.

- Absolute increase: 5,22,107 cases
- Overall increase: 8.90%
- Annual increase:
  - 2022: 2.14%
  - 2023: 1.34%
  - 2024: 2.08%
  - 2025: 3.07%

The highest annual increase occurred in 2025 (3.07%), indicating that the accumulated judicial burden continued to expand.

More High Courts experienced an overall increase than a decrease

On comparing 2021 and 2025, 15 of the 25 High Courts recorded an overall increase in pendency, while 10 High Courts recorded an overall reduction.

This demonstrates that although certain jurisdictions have succeeded in controlling their backlog, the broader High Court system continues to face substantial pendency.

Allahabad High Court has the highest pendency

The Allahabad High Court has recorded the highest pending caseload among all High Courts in the dataset over the period studied, and this shows a real challenge.

Its pendency grew from 10,24,931 cases in 2021 to 12,18,984 cases in 2025, a jump.

That is an increase of 18.93 percent, which is a lot.

This large caseload has implications for how quickly cases are decided and shows that we need better case-management strategies for courts that handle a lot of cases.

Rajasthan also witnessed growth

The Rajasthan High Court recorded an increase from 5,42,658 cases in 2021 to 6,82,725 cases in 2025, representing an overall increase of 1,40,067 cases or 25.82 percent, and the numbers are climbing fast.

The sharp increase in the final year indicates the need for stronger mechanisms to prevent further accumulation, so we must act.

Karnataka and Madhya Pradesh show growth

The Karnataka High Court increased from 2,49,451 to 3,28,599 cases, representing a 31.73 percent increase, and this is a clear signal that capacity is being stretched.

Similarly, the Madhya Pradesh High Court increased from 4,11,909 to 4,80,537 cases, representing a 16.66 percent increase. We must watch this trend closely.

Both the Karnataka High Court and the Madhya Pradesh High Court demonstrate that persistent growth in institution and/or insufficient disposal capacity can result in long-term accumulation, and this is a warning.

Some High Courts demonstrated successful reduction

The data also provides positive evidence. Several High Courts reduced their pending cases between 2021 and 2025 (Table 1):

Table 1. High Courts that reduced their pending cases between 2021 and 2025

High Court	Overall Change
Tripura	−19.31%
Jharkhand	−18.63%
Orissa	−17.71%
Jammu & Kashmir	−9.14%
Madras	−8.93%
Chhattisgarh	−6.46%
Punjab & Haryana	−6.43%
Telangana	−2.70%
Patna	−3.14%

These results establish that reduction of judicial backlog is achievable when appropriate disposal and case-management mechanisms are effectively implemented.

Madras High Court demonstrates sustained improvement

The Madras High Court shows a positive trend. Pendency fell steadily from 5,99,825 in 2021 to 5,46,285 in 2025, showing an 8.93 percent reduction.

The lack of a rise in cases for the Madras High Court shows that the backlog is being reduced in a steady way.

Orissa and Tripura show fluctuating improvement

The Orissa High Court cut pendency by 17.71 percent overall and Tripura saw a 19.31 percent reduction.

Both saw case numbers rise again in the final year. This shows that a short-term cut is not enough; we need case-management methods.

Smaller High Courts show large percentage variations

High Courts such as Sikkim and Manipur show relatively small absolute numbers but considerable percentage changes.

For example, Sikkim increased from 180 to 272 cases, representing 51.11% growth. Although this percentage is high, the absolute increase is only 92 cases.

Therefore, percentage change should always be interpreted alongside absolute pendency.

Pendency has direct implications for speedy and fair justice

The persistent increase in pending cases creates challenges for the constitutional objective of speedy justice under Article 21. Delay may affect:

- the timely resolution of disputes;
- victims seeking remedies;
- accused persons awaiting trial;
- availability and reliability of evidence;
- witness participation;
- legal costs; and
- public confidence in the system.

However, quick handling of cases must not hurt fairness. The goal should be to handle cases while still following the right steps, instead of just getting rid of the case by ignoring the rules that protect people's rights.

Conclusion

The five-year study shows that judicial pendency is still one of the challenges facing the Indian High Court system.

The total number of pending cases rose from 58,68,423 in 2021 to 63,90,530 in 2025, a rise of 8.90%.

The findings show a mixed picture. Fifteen High Courts saw a rise in pending cases, while ten High Courts managed to lower their pendency. This variation shows that judicial backlog is not the same across India and that factors such as institutional capacity, case-flow management, judicial strength, procedural practices and local workload patterns can affect pendency.

The evidence also shows that reducing pendency is possible. The good results from Jharkhand, Tripura, Orissa, Madras, Jammu & Kashmir, Punjab and Haryana and other High Courts give examples for creating better case-management policies.

At the same time, the continuing rise in total pendency shows that current measures are still not enough to reverse the national trend. The growing backlog threatens the delivery of timely justice and may weaken the idea that justice should be fair and also given in a reasonable time.

Therefore, the main conclusion of this study is that India needs a model of quick adjudication that mixes speed, fairness, judicial independence, due process and quality of decisions. Quick disposal by itself is not enough for reform if it hurts the rights of litigants.

Suggestions

Increase judicial strength

Vacancies in High Courts should be filled quickly. Having enough judges compared to the number of cases is important to stop new cases from piling up.

Strengthen case-flow management

High Courts should use ways to manage their dockets and the flow of cases, including grouping cases by how old they are, how urgent they are, how complex they are and what kind of disagreement they involve.

Old and vulnerable cases

Cases that have been waiting for a long time should be given special attention. Special groups of judges or special lists can be made for cases that have been delayed for a long time, for older people involved in legal issues, for people who are in jail without a trial and for other people who need help.

Reduce unnecessary adjournments

Unnecessary and repeated delays in court should be reduced by following the rules carefully. There should still be fair chances for people to share their side of the story.

Strengthen courts

More use of digital filing, electronic records of cases, online court hearings, sending notices through electronic means and managing evidence digitally should be encouraged, while keeping security, privacy and proper rules in place.

Promote alternative dispute resolution

Mediation, arbitration, conciliation, Lok Adalats and other ways to settle disputes without going to court should be made stronger for cases that can be settled. This can help reduce the work of courts and let judges focus on cases that need a real decision.

Establish case-management mechanisms

High Courts that handle a lot of cases, like Allahabad, Rajasthan, Bombay, Madras and Madhya Pradesh, can use special ways to manage their cases based on the type of cases they get and how many they have.

Improve court infrastructure

Judicial reform needs courtrooms, trained staff, people who take notes, researchers, technology and support from forensic experts when needed.

Strengthen data-based judicial administration

Pendency should be monitored through indicators such as:

- institution rate;
- disposal rate;
- clearance rate;
- age-wise pendency;
- average duration of cases;
- judge-wise workload; and
- category-wise pendency.

This would give a better picture than raw pendency numbers alone.

Balance speed with fairness

Any policy designed to reduce pendency must protect:

- the right to be heard;
- legal representation;
- natural justice;
- evidentiary safeguards;
- reasoned judgment; and
- timely disposal.

Thus, fast justice should not mean rushed justice; it should mean justice without losing fairness.

Develop a framework for expedited adjudication

A coordinated framework involving the Supreme Court, High Courts, government, judiciary and legal-services institutions could set clear targets to cut backlog while keeping judicial independence and case complexity in mind.

Regular quinquennial empirical assessment

The Government and judiciary should publish clear year-by-year pendency data and carry out periodic studies. These studies can spot good practices and let them be copied in other areas.

Implications for Judicial Reform

The empirical results suggest the need for a strategy involving:

- timely filling of judicial vacancies;
- strengthening court infrastructure and staff;
- structured case-flow and docket management;
- prioritising old and long-pending cases;
- cutting unnecessary adjournments;
- using more digital case-management systems;
- improving e-Courts and virtual hearings;
- promoting mediation and other ADR mechanisms;
- better coordination between investigation, prosecution and courts in criminal matters; and
- regular monitoring of institution, disposal and pendency rates.

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RESEARCH ARTICLE · HEALTH AND NURSING SCIENCES

A STUDY TO ASSESS THE LEVEL OF STRESS AMONG B.SC. NURSING STUDENTS IN A SELECTED NURSING INSTITUTE OF GUWAHATI, ASSAM

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KEYWORDS

Stress, Perceived Stress, PSS-10, B.Sc. Nursing Students, Moderate Stress, Clinical Education

ABSTRACT

Nursing education is a challenging and demanding professional course that requires students to acquire theoretical knowledge, develop clinical skills, and demonstrate appropriate attitudes and professional competencies. B.Sc. Nursing students are exposed to various academic and clinical responsibilities throughout their course. As they progress towards the final semester, they are expected to manage increased academic workload, clinical duties, assignments, examinations, practical requirements, and preparation for their future professional roles.

Assessing the level of stress among nursing students is therefore important for identifying students who may require appropriate guidance and support. Understanding the extent of stress can help nursing educators and institutions develop suitable stress-management strategies, counselling services, supportive learning environments, and educational interventions to promote students' mental well-being and academic success.

Therefore, I intend to conduct a study entitled "A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam". The study aims to assess the level of stress among the students and provide useful information for planning appropriate measures to help them cope effectively with academic and clinical stress.

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Introduction

Stress is a common psychological response that occurs when an individual perceives that the demands placed upon them exceed their ability to cope[1]. A certain amount of stress may motivate students to perform better; however, excessive or prolonged stress can negatively affect their physical, psychological, social, and academic well-being[2]. Nursing students may experience stress due to heavy academic workload, clinical responsibilities, fear of examinations, time constraints, interpersonal relationships, lack of sleep, financial concerns, and uncertainty regarding their future career.

At this stage, students may experience additional pressure related to final examinations, clinical performance, research work, completion of academic requirements, and expectations regarding employment and professional responsibilities[3]. If stress is not appropriately recognized and managed, it may affect concentration, academic performance, confidence, clinical competence, and overall quality of life.

Background of the study

The present study is undertaken to assess the level of stress among B.Sc. Nursing students of a selected nursing institute, Guwahati, Assam. Nursing education is a demanding professional programme in which students are required to manage theoretical learning, clinical practice, assignments, examinations, research activities, and patient-care responsibilities.

Stress is an important concern among nursing students because continuous academic and clinical demands may affect their physical, psychological, social, and academic well-being. Students in the 8th semester may experience additional stress as they approach the completion of their course and prepare for final examinations, professional responsibilities, employment, and transition from student to staff nurse.

The purpose of this study is to determine the level of stress experienced by B.Sc. Nursing students and to identify the extent to which stress is present among them. The study will provide useful information regarding the stress experienced by students during this important stage of their nursing education.

The study will be conducted among B.Sc. Nursing students of a selected institute, Guwahati, Assam. A suitable standardized or structured stress assessment tool will be used to assess the students' level of stress. The collected data will be systematically analysed and interpreted according to the objectives of the study.

The findings of the study may help nursing teachers and administrators understand the stress experienced by students and plan appropriate measures such as counselling, stress-management education, relaxation techniques, time-management strategies, peer support, and other supportive interventions. Ultimately, the study may contribute to improving the psychological well-being, academic performance, clinical learning, and professional preparedness of B.Sc. Nursing students.

Need of the study

Nursing is a profession that requires students to develop sound theoretical knowledge, clinical competence, communication skills, and professional responsibility. During their nursing education, students are exposed to various academic and clinical demands that may contribute to stress. Stress is a natural response to challenging situations, but excessive or prolonged stress may adversely affect students' physical health, psychological well-being, academic performance, and clinical learning[4].

B.Sc. Nursing students face several stress-producing situations, such as heavy academic workload, examinations, assignments, clinical postings, patient-care responsibilities, time constraints, long working hours, lack of adequate rest, fear of making mistakes, and difficulties in balancing academic and personal responsibilities. These challenges may become more significant as students near the completion of their course and face final examinations, clinical responsibilities, research work, and preparation for their future professional career[5].

The pressure to perform well academically and clinically may increase feelings of anxiety, tension, frustration, and emotional exhaustion. High levels of stress may result in poor concentration, disturbed sleep, reduced motivation, decreased academic performance, and difficulty in providing effective patient care.

Assessing stress among nursing students is important because early identification of stress can help educators understand the problems faced by students and provide appropriate support. Counselling, stress-management programmes, relaxation techniques, time-management education, peer support, and a supportive academic and clinical environment may help students cope more effectively with stress.

Therefore, I (the researcher) felt the need to conduct a study entitled "A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam"[6]. The findings of the study may help nursing educators and administrators understand the level of stress among students and plan appropriate measures to promote their psychological well-being, academic achievement, clinical competence, and successful transition into professional nursing practice.

Problem statement

"A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam"

Objectives of the study

1. To assess the level of stress among the B.Sc. Nursing students in the selected nursing institute.
2. To determine the association between stress levels and selected socio-demographic variables such as age, residence, family type, previous academic performance, number of family members, birth order, history of academic failure, parents' education and current financial support.

Operational definitions

1. **Stress:** The physical, emotional, and psychological response experienced by 8th-semester B.Sc. Nursing students due to academic, personal, and environmental factors, measured by a standardized stress scale, that is, the stress assessment tool Perceived Stress Scale (PSS-10).
2. **8th-semester B.Sc. Nursing students:** Students studying in the 8th semester of the B.Sc. Nursing programme at Regional College of Nursing, Guwahati, Assam, under SSUHS.

Assumptions

For the research topic "A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam", the following assumptions were made:

1. B.Sc. Nursing students experience stress during their course in the selected nursing institute of Guwahati,

Assam.

2. Academic and personal factors contribute to stress levels in the selected nursing institute of Guwahati, Assam.
3. Students will provide honest responses in the selected nursing institute of Guwahati, Assam.

Research hypotheses

1. **H1:** There is a significant level of stress among B.Sc. Nursing students.
2. **H2:** There is a significant association between stress levels and selected socio-demographic variables.

Delimitations

For the research topic "A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam", the following delimitations were set:

1. The study will be limited to B.Sc. Nursing students.
2. The study will be conducted only at Regional College of Nursing, Guwahati, Assam.
3. Data collection will be limited to the selected period.

Review of Literature

For the study "A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam", the review of literature is organized into the following sections[7].

Literature related to stress among nursing students

Nursing education involves academic learning, clinical practice, examinations, assignments, patient-care responsibilities and professional skill development. These demands can expose nursing students to considerable psychological stress.

A systematic review and analysis published in 2022 examined stress among nursing students using validated stress-assessment instruments. The review included studies involving 9,202 participants and reported that stress among nursing students was predominantly at a moderate level. The Perceived Stress Scale (PSS) was among the commonly used instruments in the included studies[8].

Literature related to perceived stress among undergraduate nursing students

A descriptive cross-sectional study conducted among 282 undergraduate nursing students in North India assessed perceived stress using the Perceived Stress Scale. The study reported a mean perceived stress score of 28.67 ± 5.32. Female students had a higher mean perceived stress score than male students, and differences were also observed across academic years. The researchers concluded that nursing students experienced considerable psychological stress and recommended measures to reduce stress among students.

This study is particularly relevant to the present research because it was conducted in India and used the Perceived Stress Scale, which is also the instrument used in the present study.

Literature related to clinical stress among nursing students

Clinical practice is an important component of nursing education but can also be a significant source of stress. A systematic review published in 2023 examined the relationship between perceived clinical stress and clinical performance among prelicensure nursing students. The review identified gender, academic year and clinical skill preparation as important themes and noted that sustained clinical stress may affect students' perceptions of competence, confidence and clinical performance.

A 2021 instrument-development study involving 430 nursing students identified several important clinical stressors, including inadequate knowledge and skills, inappropriate clinical environments, limitations in clinical instruction, inappropriate instructor conduct and concerns regarding nursing as a career. The findings demonstrate that stress among nursing students can arise from multiple academic and clinical factors rather than from a single source.

Literature related to sources of stress and coping

An umbrella review published in 2024 concluded that prelicensure nursing students experience significant stress associated with both their educational programme and clinical placements. The review highlighted the importance of effective coping mechanisms to help students manage academic and clinical responsibilities and to support their well-being and academic performance.

Literature related to stress assessment using PSS-10

The Perceived Stress Scale (PSS) is widely used to assess the degree to which individuals perceive situations in their lives as stressful. Recent systematic-review evidence shows that the PSS-10 is one of the commonly used instruments for assessing stress among undergraduate nursing students. A 2026 systematic review identified PSS-based studies as an important group of studies in the nursing-student stress literature.

Research gap

The literature indicates that stress is a significant concern among nursing students and may be associated with

academic demands, clinical experiences, preparation for professional practice and other personal or contextual factors. However, stress levels can differ according to academic year, institution, clinical environment, geographical location and measurement instrument.

Therefore, there is a need to assess perceived stress specifically among B.Sc. Nursing students at Regional College of Nursing, Guwahati, Assam. The present study has helped determine the level of perceived stress among students at this particular stage of nursing education and may provide a basis for appropriate stress-management and supportive interventions.

Summary of review of literature and relevance to the present study

1. **Stress among nursing students:** Nursing students experience substantial stress — establishes the importance of the problem.
2. **Perceived stress in Indian nursing students:** High perceived stress was reported among undergraduate nursing students in North India — provides Indian evidence.
3. **Clinical stress:** Clinical environment, skills and academic year influence stress — relevant to nursing students.
4. **Sources and coping:** Academic and clinical responsibilities are major stressors — supports the need for stress-management measures.
5. **PSS-10 assessment:** PSS is commonly used to measure perceived stress — supports the selection of PSS-10 for the study.
6. **Research gap:** Stress varies by context and academic stage.

Study Overview

Background

Stress is an important concern among nursing students because of the multiple academic, clinical and personal demands associated with nursing education. Students are required to manage theoretical learning, clinical postings, practical examinations, assignments, patient-care responsibilities and preparation for professional practice. The final semester may be particularly demanding because students are approaching completion of their course and transition into professional nursing practice. Persistent or excessive stress may negatively affect students' concentration, academic performance, clinical competence, psychological well-being and overall quality of life. Therefore, assessment of perceived stress among final-semester nursing students is important for identifying students who may require appropriate support and stress-management interventions.

Aim of the study

The aim of the study was to assess the level of perceived stress among B.Sc. Nursing students in a selected nursing institute of Guwahati, Assam.

Objectives

1. To assess the level of stress among B.Sc. Nursing students in the selected nursing institute of Guwahati, Assam.
2. To describe the selected demographic characteristics of the study participants in the selected nursing institute of Guwahati, Assam.
3. To determine the distribution of students according to low, moderate and high levels of perceived stress in the selected nursing institute of Guwahati, Assam.
4. To calculate the mean and standard deviation of the PSS-10 scores among the study participants in the selected nursing institute of Guwahati, Assam.

Methodology

A quantitative descriptive research approach was adopted for the study "A Study to Assess the Level of Stress among B.Sc. Nursing Students in a Selected Nursing Institute of Guwahati, Assam".

The participants were assessed using a structured socio-demographic questionnaire and the 10-item Perceived Stress Scale (PSS-10).

The PSS-10 was used to measure the students' perceived stress. The total score ranges from 0 to 40, with higher scores indicating greater perceived stress. For the present analysis, scores were classified into low, moderate and high stress categories. The collected data were organized and analyzed using frequency, percentage, mean, median and standard deviation.

Results

The socio-demographic findings showed that the largest proportion of students were 23 years old (42.22%), 48.89% were from urban areas, and 84.44% belonged to nuclear families. About 62.22% had 4–6 family members. Regarding academic characteristics, 80% of students had good academic performance and 80% had no history of academic failure or backlog. Parents/family were the source of financial support for 97.78% of the participants.

The PSS-10 analysis revealed that 8 students (17.78%) had low stress, 35 students (77.78%) had moderate stress, and 2 students (4.44%) had high stress. The overall mean PSS-10 score was 19.16 with a standard deviation of 6.31,

and the median score was 21. Thus, moderate stress was the predominant level among the study participants.

Research Methodology

Research approach: A quantitative research approach was adopted for the present study. This approach is appropriate because the study aims to objectively assess and quantify the level of stress among B.Sc. Nursing students[9].

Research design: A descriptive research design was used to assess the level of stress among B.Sc. Nursing students.

Setting of the study: The study was conducted at Regional College of Nursing, Guwahati, Assam. The college provides undergraduate/postgraduate nursing education and clinical learning opportunities.

Population: The target population consisted of B.Sc. Nursing students studying at Regional College of Nursing, Guwahati, Assam[10].

Sample: The sample comprised B.Sc. Nursing students who fulfilled the predetermined inclusion criteria and were available during the period of data collection.

Sample size: The sample size was determined according to the number of eligible B.Sc. Nursing students available in the selected setting and the requirements of the study. It was found to be 45.

Sampling technique: An appropriate sampling technique, such as total enumeration sampling or convenience sampling, was used to select the participants depending on the availability and eligibility of the students.

Inclusion criteria

Students were included in the study if they:

- were studying in the 8th semester of B.Sc. Nursing at Regional College of Nursing, Guwahati, under SSUHS[11];
- were willing to participate in the study; and
- were available during the period of data collection.

Exclusion criteria

Students were excluded if they:

- were absent during data collection.

Variables of the study

- **Research variable:** Level of stress in B.Sc. Nursing students at Regional College of Nursing, Guwahati, Assam.
- **Selected socio-demographic variables:** Age, residence, previous academic background, type of family, number of family members, birth order, history of academic failure, parents' education and current financial support.

Data collection tool

Data were collected using two sections:

- **Section A: Socio-demographic variables.** Socio-demographic variables consist of age, residence, previous academic background, type of family, number of family members, birth order, history of academic failure, parents' education and current financial support.
- **Section B: Stress assessment scale (PSS-10).** A standardized stress assessment scale (PSS-10) was used to assess the level of stress among the students. The scoring and interpretation were carried out according to the instructions of the selected scale.

Validity and reliability

The standard stress assessment scale PSS-10 was used for stress assessment.

The reliability of the tool was established using an appropriate reliability-testing method before the final data collection. The reliability of the tool is 0.79, and it is considered to be reliable and adequate.

Data collection procedure

After obtaining permission from the concerned authority and informed consent from the participants, the researcher collected data from eligible B.Sc. Nursing students. The purpose of the study was explained to the participants. The questionnaire/stress assessment scale was administered, and sufficient time was provided for completion. Confidentiality and anonymity of the participants were maintained.

Ethical considerations

1. The study was done after obtaining ethical approval from the Institutional Ethical Committee, Regional College of Nursing, Guwahati.
2. Verbal and written consent was obtained from all the participants of the study after explaining the purpose and other details of the study.
3. The participants were assured of the confidentiality of the data obtained and that participation was voluntary.

Plan for data analysis

The collected data were organized, coded, tabulated, and analysed using descriptive and inferential statistics.

- **Descriptive statistics:** Frequency and percentage for socio-demographic variables. Mean and standard deviation for stress scores. Frequency and percentage for different levels of stress.

- **Inferential statistics:** Appropriate statistical tests, such as the chi-square test, were used to determine the association between the level of stress and selected socio-demographic variables, according to the nature of the data.

Expected outcome

The study determined the level of stress among B.Sc. Nursing students and identified whether the stress level was low, moderate, or high according to the scoring criteria of the selected assessment tool. The findings may help nursing educators and administrators develop appropriate stress-management and supportive interventions for students.

Schematic diagram

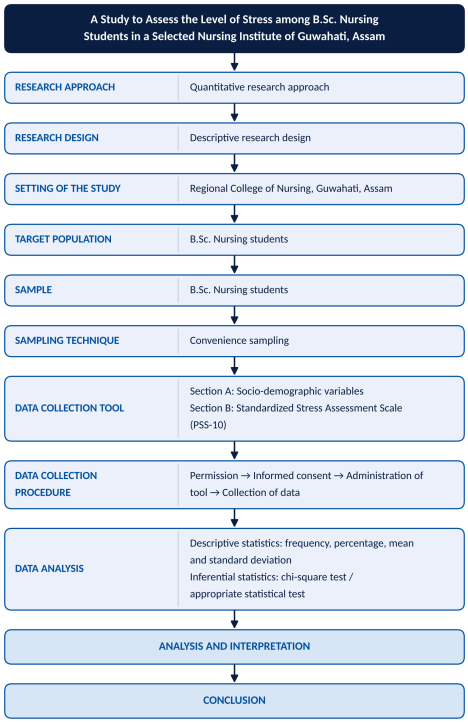


Figure 1. Schematic diagram of the research methodology of the study

Analysis and Interpretation

Socio-demographic data

Table 1. Distribution of students according to age (N = 45)

Age	Frequency (f)	Percentage (%)
22 years	10	22.22
23 years	19	42.22
24 years	10	22.22
25 years	5	11.11
26 years	1	2.22
Total	45	100

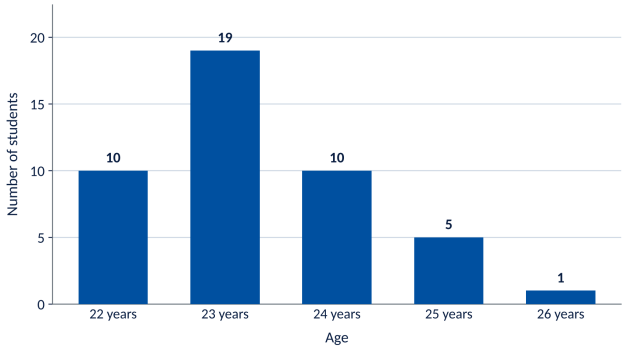


Figure 2. Age distribution of 8th-semester B.Sc. Nursing students (N = 45)

Interpretation: The majority of students, 19 (42.22%), were 23 years old. The least number, 1 (2.22%), were 26 years old.

Table 2. Distribution of students according to place of residence (N = 45)

Place of residence	Frequency (f)	Percentage (%)
Urban	22	48.89
Rural	16	35.56
Semi-urban	7	15.56
Total	45	100

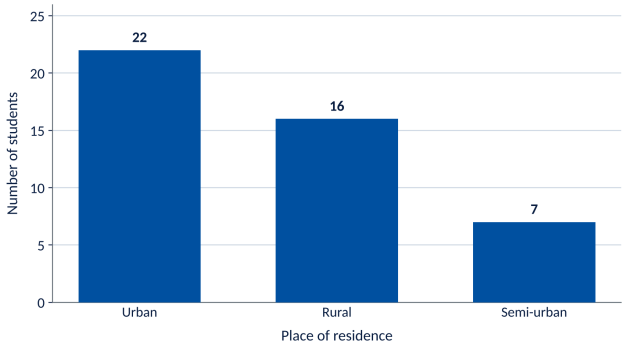


Figure 3. Distribution of students according to place of residence (N = 45)

Interpretation: Nearly half of the students, 22 (48.89%), were from urban areas, while 7 (15.56%) were from semi-urban areas.

Table 3. Distribution of students according to type of family (N = 45)

Type of family	Frequency (f)	Percentage (%)
Nuclear	38	84.44
Joint	6	13.33
Extended	1	2.22
Total	45	100

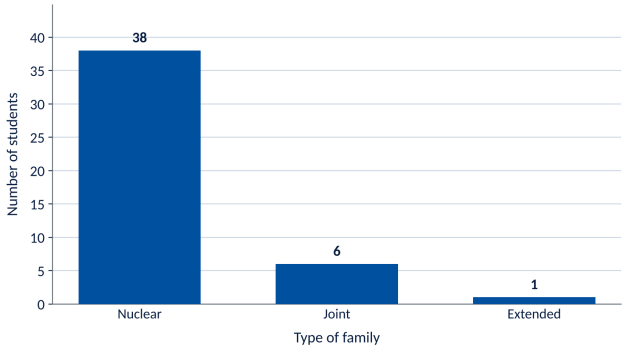


Figure 4. Distribution of students according to type of family (N = 45)

Interpretation: The majority, 38 (84.44%), belonged to nuclear families.

Table 4. Distribution of students according to number of family members (N = 45)

Number of family members	Frequency (f)	Percentage (%)
1–3	8	17.78
4–6	28	62.22
7 or more	9	20.00
Total	45	100

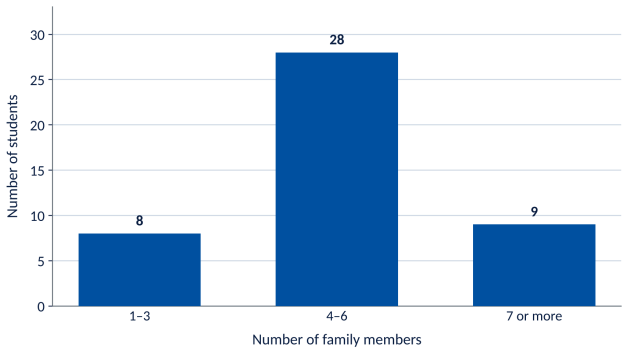


Figure 5. Distribution of students according to number of family members (N = 45)

Interpretation: Most students, 28 (62.22%), had 4–6 members in their family.

Table 5. Distribution of students according to birth order (N = 45)

Birth order	Frequency (f)	Percentage (%)
First	20	44.44
Second	18	40.00
Third	2	4.44
Fourth or above	5	11.11
Total	45	100

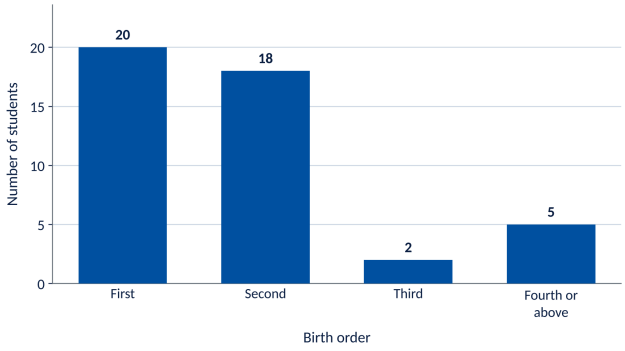


Figure 6. Distribution of students according to birth order (N = 45)

Interpretation: The highest proportion, 20 (44.44%), were first-born children, followed by 18 (40.00%) who were second-born.

Table 6. Distribution of students according to academic performance in previous semester(s) (N = 45)

Academic performance	Frequency (f)	Percentage (%)
Excellent	5	11.11
Good	36	80.00
Average	4	8.89
Below average	0	0
Total	45	100

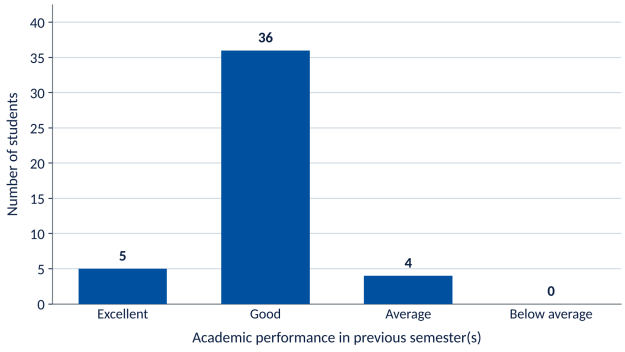


Figure 7. Distribution of students according to academic performance in previous semester(s) (N = 45)

Interpretation: The majority of students, 36 (80.00%), had good academic performance in their previous semester(s). None of the students were categorized as below average.

Table 7. Distribution of students according to history of academic failure/backlog (N = 45)

Academic failure/backlog	Frequency (f)	Percentage (%)
No	36	80.00
Yes	9	20.00
Total	45	100

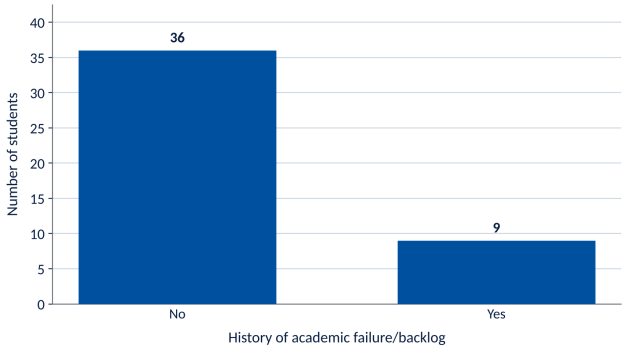


Figure 8. Distribution of students according to history of academic failure/backlog (N = 45)

Interpretation: Most students, 36 (80.00%), reported no history of academic failure/backlog, whereas 9 (20.00%) reported a history of academic failure/backlog.

Table 8. Distribution of students according to father's educational status (N = 45)

Father's educational status	Frequency (f)	Percentage (%)
No formal education	0	0
Primary education	1	2.22
Secondary education	9	20.00
Higher secondary education	9	20.00
Graduate	22	48.89
Postgraduate and above	4	8.89
Total	45	100

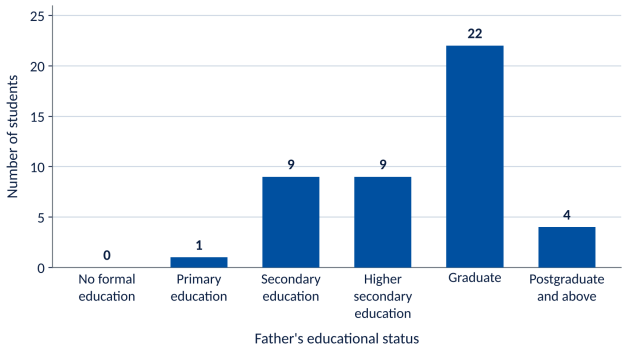


Figure 9. Distribution of students according to father's educational status (N = 45)

Interpretation: Nearly half of the students, 22 (48.89%), had fathers who were graduates. No father was recorded under no formal education.

Table 9. Distribution of students according to mother's educational status (N = 45)

Mother's educational status	Frequency (f)	Percentage (%)
No formal education	1	2.22
Primary education	3	6.67
Secondary education	8	17.78
Higher secondary education	20	44.44
Graduate	12	26.67
Postgraduate and above	1	2.22
Total	45	100

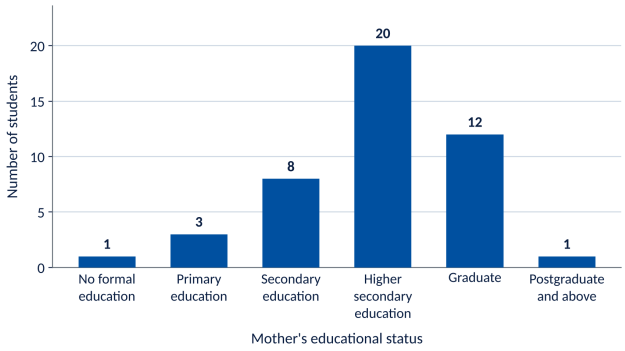


Figure 10. Distribution of students according to mother's educational status (N = 45)

Interpretation: The largest proportion, 20 (44.44%), had mothers with higher secondary education, followed by 12 (26.67%) whose mothers were graduates.

Table 10. Distribution of students according to current financial support for education (N = 45)

Source of financial support	Frequency (f)	Percentage (%)
Parents/family	44	97.78
Self	0	0
Scholarship	1	2.22
Educational loan	0	0
Other	0	0
Total	45	100

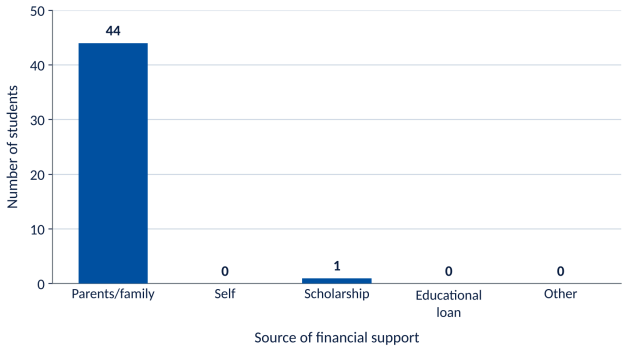


Figure 11. Distribution of students according to source of financial support for education (N = 45)

Interpretation: All 45 (100%) students reported that their education was financially supported by their parents/family.

Overall socio-demographic findings

The socio-demographic analysis of the 45 8th-semester B.Sc. Nursing students shows that the largest proportion were 23 years old (42.22%), lived in urban areas (48.89%), belonged to nuclear families (84.44%), had 4–6 family members (62.22%), were first-born (44.44%), and had good previous academic performance (80.00%). Most students had no history of academic failure/backlog (80.00%). Regarding parental education, fathers were most commonly graduates (48.89%), while mothers most commonly had higher secondary education (44.44%). All students reported parents/family as their source of financial support (100%).

PSS-10 analysis

The PSS-10 was used to assess perceived stress. The total score ranges from 0–40. Higher scores indicate greater perceived stress.

Table 11. Distribution of students according to level of perceived stress (N = 45)[12]

Level of stress	PSS-10 score	Frequency (f)	Percentage (%)
Low	0–13	6	13.33
Moderate	14–26	33	73.33
High	27–40	6	13.33
Total		45	100

Note: Mean = 19.16; SD = 5.65.

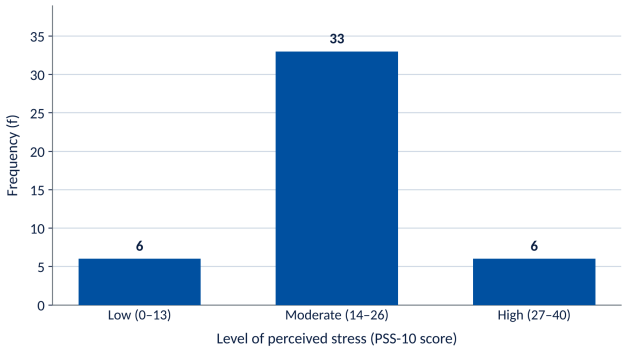


Figure 12. Level of perceived stress among B.Sc. Nursing students: distribution of 45 students according to PSS-10 stress category (low 0–13, moderate 14–26, high 27–40; N = 45)

Table 12. Percentage distribution of students according to PSS-10 stress level (N = 45)

Low	Moderate	High
13.33%	73.33%	13.33%

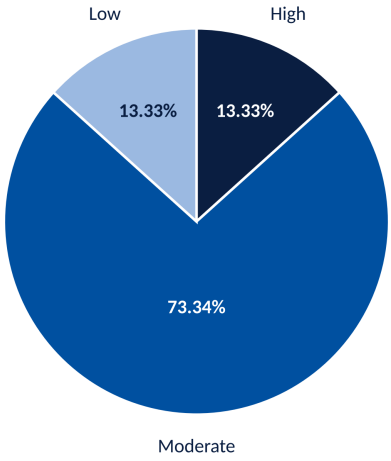


Figure 13. Percentage distribution of students according to PSS-10 stress level (N = 45)

Interpretation: Table 11 and Figures 12 and 13 show that out of 45 B.Sc. Nursing students, the majority, 33 (73.33%), experienced moderate perceived stress. 6 (13.33%) students had low perceived stress, while 6 (13.33%) students had high perceived stress.

The findings indicate that moderate stress was the predominant level of perceived stress among the students.

Descriptive statistics of PSS-10

Table 13. Descriptive statistics of PSS-10 scores

Number of students	Minimum score	Maximum score	Mean score	Median score
45	4	30	19.16	19

Discussion

The present study was conducted to assess the level of stress among B.Sc. Nursing students in a selected nursing institute of Guwahati, Assam. A total of 45 students were included in the study. The data were analyzed using frequency, percentage, and descriptive statistics. The Perceived Stress Scale (PSS-10) was used to assess the level of perceived stress.

Discussion of demographic data

The demographic analysis showed that the largest proportion of students were 23 years old (42.22%). Students aged 22 and 24 years constituted 22.22% each, while 11.11% were 25 years old and 2.22% were 26 years

old. Regarding place of residence, 48.89% of students were from urban areas, 35.56% were from rural areas and 15.56% were from semi-urban areas. Thus, urban students constituted the largest group.

With regard to type of family, most students (84.44%) belonged to nuclear families. Joint families accounted for 13.33%, while only 2.22% belonged to extended families. In terms of family size, the majority (62.22%) had 4–6 family members.

Regarding birth order, 44.44% of the students were first-born, 40% were second-born, 4.44% were third-born and 11.11% were fourth-born or above. The analysis of academic performance revealed that 80% of students had good academic performance, 11.11% had excellent performance and 8.89% had average performance. No student was categorized as below average. Similarly, 80% reported no history of academic failure or backlog, whereas 20% reported a previous academic failure/backlog.

Regarding father's educational status, the highest proportion (48.89%) had graduate fathers. In relation to mother's educational status, the largest proportion (44.44%) had mothers with higher secondary education.

97.78% of students reported that their parents/family provided financial support for their education.

Discussion of PSS-10 data

The PSS-10 analysis revealed that moderate stress was the predominant level of perceived stress among the students.

Out of 45 students:

- 6 (13.33%) had low stress;
- 33 (73.33%) had moderate stress; and
- 6 (13.33%) had high stress.

The mean PSS-10 score was 19.16, with a median of 19. The scores ranged from 4 to 30. The mean score falls within the moderate-stress category, indicating that the study group as a whole experienced a moderate level of perceived stress. The finding that nearly three-fourths of the students experienced moderate stress may be related to the academic and clinical demands associated with the final semester of nursing education. Students have to manage theoretical studies, clinical postings, practical requirements, examinations, assignments and preparation for professional responsibilities simultaneously.

The presence of 6 students (13.33%) with high stress is noteworthy. Although they represented a smaller proportion of the sample, these students may require additional attention and appropriate support. Conversely, 6 students (13.33%) had low stress, suggesting that students differ in their responses and coping abilities despite being exposed to similar educational circumstances.

Overall discussion

The overall data analysis demonstrates that moderate perceived stress is prevalent among 8th-semester B.Sc. Nursing students. The findings emphasize the importance of providing students with appropriate stress-management strategies, counselling, peer support, time-management skills and opportunities to discuss academic and clinical difficulties.

The demographic findings provide useful background information about the study participants, while the PSS-10 findings directly address the primary objective of the research. However, the present frequency and percentage analysis does not establish a statistically significant relationship between demographic variables and stress. Such relationships would require inferential statistical tests such as the chi-square test.

Summary of the main finding

The majority of B.Sc. Nursing students (73.33%) experienced moderate perceived stress, while 13.33% experienced low stress and 13.33% experienced high stress. The overall mean PSS-10 score was 19.16, indicating moderate perceived stress among the study participants.

This finding supports the need for early identification and management of stress among nursing students, particularly during the final year when academic and clinical responsibilities are substantial.

Summary and Conclusion

The present study was conducted to assess the level of stress among 8th-semester B.Sc. Nursing students of Regional College of Nursing, Guwahati. A quantitative descriptive research approach was adopted, and data were collected from 45 students using a demographic questionnaire and the Perceived Stress Scale (PSS-10).

The demographic analysis showed that most students were 23 years old (42.22%), belonged to nuclear families (84.44%), were from urban areas (48.89%), and had 4–6 family members (62.22%). Most students had good academic performance (80%) and no history of academic failure or backlog (80%). Parents or family members were the source of financial support for education for 100% of the students.

The PSS-10 analysis revealed that 33 students (73.33%) experienced moderate stress, while 6 students (13.33%) experienced low stress and 6 students (13.33%) experienced high stress. The overall mean PSS-10 score was 19.16, with a median score of 19. The findings

indicate that moderate stress was the predominant level of perceived stress among the study participants.

In conclusion, the study demonstrates that stress is an important concern among 8th-semester B.Sc. Nursing students. Although the majority experienced moderate stress, the presence of students with high stress highlights the need for early identification and appropriate support. Nursing institutions should provide counselling, stress-management education, relaxation techniques, time-management training, peer support and academic/clinical guidance to help students cope effectively with stress.

Overall, the study concludes that appropriate preventive and supportive measures are necessary to promote the psychological well-being, academic performance and clinical competence of nursing students.

Conclusion

The study concluded that the majority of B.Sc. Nursing students experienced moderate perceived stress. Although most students had good academic performance and no history of academic backlog, a substantial proportion experienced moderate stress, indicating that the demands of nursing education may contribute to perceived stress.

The findings emphasize the need for regular stress assessment, counselling, stress-management education, relaxation techniques, time-management training, peer support and appropriate academic and clinical guidance. Such interventions may help nursing students develop effective coping strategies, maintain psychological well-being and improve their academic and clinical performance.

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RESEARCH ARTICLE · HEALTH AND NURSING SCIENCES

A STUDY TO ASSESS KNOWLEDGE, SKILLS AND PRACTICE REGARDING PATIENT SAFETY AMONG B.SC. NURSING STUDENTS OF REGIONAL COLLEGE OF NURSING, GUWAHATI, ASSAM

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KEYWORDS

Patient Safety, Knowledge, Skills, Practice, B.Sc. Nursing Students, Clinical Practice, Nursing Education

ABSTRACT

Background: Patient safety is an essential component of quality nursing care. Nursing students need adequate knowledge, appropriate clinical skills, and safe practices to prevent errors and avoidable harm to patients. Assessing these competencies during undergraduate nursing education can help identify areas requiring improvement.

Objective: To assess the knowledge, skills, and practice regarding patient safety among 2nd-semester B.Sc. Nursing students of Regional College of Nursing, Guwahati, Assam.

Methodology: A quantitative descriptive research approach was adopted. The study was conducted among 42 second-semester B.Sc. Nursing students. Data were collected using appropriate structured assessment tools to assess knowledge, skills, and practice related to patient safety. The collected data were organized, analyzed, and presented using descriptive statistics such as frequency, percentage, and graphical representation.

Results: The findings showed variation in the levels of knowledge, skills, and practice regarding patient safety among the students. The assessment highlighted the importance of strengthening the application of patient-safety principles during clinical practice. The findings also indicated the need for continuous reinforcement through classroom teaching, demonstrations, simulation, supervised clinical practice, and feedback.

Conclusion: The study concludes that patient-safety education should be emphasized throughout undergraduate nursing education. Developing adequate knowledge along with competent clinical skills and consistent safe practices can help nursing students provide safer and higher-quality patient care.

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Introduction

Patient safety is a fundamental component of quality healthcare and an essential responsibility of every healthcare professional. It refers to the prevention of avoidable harm to patients during the process of healthcare delivery. Safe patient care requires accurate assessment, effective communication, proper infection prevention, safe medication administration, correct patient identification, prevention of falls, appropriate documentation, and timely reporting of errors or adverse events.

Nursing education provides students with opportunities to learn patient-safety principles in the classroom and apply them during clinical practice. However, students may face challenges in transferring theoretical knowledge into clinical practice because of limited experience, unfamiliar clinical environments, communication difficulties, workload, and inadequate supervision. Regular assessment of students' knowledge, skills, and practices can help identify areas requiring improvement[1].

Patient-safety education is particularly important for second-semester B.Sc. Nursing students, as they are beginning to develop and apply fundamental clinical

competencies. Early identification of gaps in patient-safety knowledge and practice can help nursing educators provide appropriate teaching, demonstration, supervision, and reinforcement.

Therefore, the present study was undertaken to assess patient safety among 42 second-semester B.Sc. Nursing students of Regional College of Nursing, Guwahati, Assam, with emphasis on their knowledge, skills, and practice related to patient safety. The study aims to provide useful information for strengthening patient-safety education and promoting safe and quality nursing care.

Need for the study

Patient safety is an essential component of quality nursing care. Nurses are directly involved in many activities that can affect patient safety, including medication administration, infection prevention, patient identification, communication, documentation, prevention of falls, and monitoring of patients. Therefore, nursing students need to develop appropriate knowledge, skills and attitudes related to patient safety before becoming independent practitioners[2].

The World Health Organization (WHO) recommends that patient safety should be incorporated into the education of nursing and other health-profession students. Its patient-safety curriculum includes areas such as teamwork, learning from errors, clinical-risk management, infection prevention and control, invasive procedures and medication safety.

Nursing students are exposed to real clinical situations during their training. Although theoretical knowledge is important, students must also be able to apply safety principles correctly during clinical practice. Previous evidence indicates that the clinical learning environment, active participation, supervision and feedback can influence the development of patient-safety competencies among undergraduate nursing students.

Patient-safety education is particularly important for students in the early semesters of B.Sc. Nursing because they are beginning to develop fundamental clinical skills and professional behaviours. Early identification of gaps in knowledge and practice can allow nursing educators to provide additional demonstrations, supervised practice, simulation and corrective feedback before students take on greater clinical responsibilities.

Patient-safety competencies also involve more than simply knowing the correct procedure. Students need to understand how communication, teamwork, clinical systems and human factors can contribute to or prevent patient harm. Current evidence continues to emphasize identifying gaps in patient-safety competencies among nursing students and strengthening nursing education accordingly.

Therefore, assessing patient safety among B.Sc. Nursing students is important for determining whether students possess adequate knowledge and are able to apply safety principles during clinical practice. The findings can help nursing educators identify areas of weakness and plan appropriate educational interventions.

Problem statement

“A study to assess the knowledge, skills and practice regarding patient safety in clinical practices among 2nd semester B.Sc. Nursing students of Regional College of Nursing, Guwahati, Assam.”

Assumptions

The study is based on the following assumptions:

1. It is assumed that B.Sc. Nursing students have some basic knowledge and awareness regarding patient safety and safe clinical practices.
2. It is assumed that nursing students’ patient-safety practices may vary according to their knowledge, clinical exposure, training, and experience.
3. It is assumed that regular clinical supervision and guidance by nursing faculty and clinical staff can improve patient-safety practices among nursing students.
4. It is assumed that assessment of patient-safety practices will help identify areas requiring further education, training, and supervision among nursing students.
5. It is assumed that improving patient-safety practices among nursing students will contribute to the delivery of safe, effective, and quality nursing care.

Delimitation

The study is delimited to government nursing colleges of Assam.

Review of Literature

Introduction

A review of literature is a systematic examination of published and available information related to the research problem. It helps the researcher understand the existing knowledge, identify gaps in previous studies and provide a scientific basis for the present investigation.

Patient safety is an essential component of quality nursing care. Nursing students need adequate knowledge, clinical skills, communication abilities and appropriate safety attitudes to provide safe care. Patient-safety education

during undergraduate nursing education can help students recognize risks, prevent errors and develop safe clinical practices.

Literature related to patient safety

The World Health Organization (WHO) developed the Multi-professional Patient Safety Curriculum Guide to support patient-safety education in universities and health-professional schools, including nursing. The guide includes major areas such as understanding human factors, teamwork, learning from errors, clinical-risk management, quality improvement, patient engagement, infection prevention, invasive procedures and medication safety.

Patient safety is therefore not limited to preventing individual errors; it also involves understanding health-care systems, communication, teamwork, clinical risks and organizational factors. Incorporating these concepts into undergraduate nursing education is important because students develop their professional behaviours and clinical competencies during their training.

Literature related to patient-safety education among nursing students

Farokhzadian et al. (2024) conducted a cross-sectional study among 215 nursing students in Iran to assess patient-safety competencies in classroom and clinical settings. The researchers used the Health Professional Education in Patient Safety Survey. The mean overall scores were 3.50 ± 0.55 in classroom settings and 3.45 ± 0.57 in clinical settings. Clinical safety had the highest scores in both settings. However, students reported lower confidence in some areas of safety culture and understanding human and environmental factors. The researchers concluded that existing educational programmes provide opportunities for developing patient-safety competencies, but additional teaching strategies and clinical approaches are required.

The same study found that approximately 84.65% of students agreed or strongly agreed that they had learned about patient-safety competencies in the classroom, while 83.25% reported this for clinical settings. The findings demonstrate that both classroom teaching and clinical exposure contribute to patient-safety learning.

Literature related to the gap between knowledge and clinical practice

Previous literature indicates that students may feel more confident about theoretical or clinical-safety aspects than about the broader sociocultural components of patient safety. A review discussed by Farokhzadian et al. identified concerns regarding the limited integration of areas such as human factors and patient-safety concepts into nursing curricula. The literature also emphasizes that the quality of the clinical learning environment can influence students' overall competence.

This is particularly important for second-semester nursing students because they are at an early stage of clinical education. Students may know the correct procedure theoretically but require repeated demonstration, return-demonstration, supervision and feedback before consistently performing it safely.

Literature related to medication safety

Medication safety is an important component of patient safety in nursing practice. Nursing students should learn correct patient identification, medication rights, checking medication orders, preventing medication errors and reporting errors or near misses.

The WHO patient-safety curriculum specifically identifies improving medication safety as one of its 11 teaching topics. It also emphasizes learning from errors and clinical-risk management as important components of patient-safety education.

Literature related to infection prevention and control

Infection prevention and control is another major component of patient safety. Appropriate hand hygiene, use of personal protective equipment, safe handling of biomedical waste and prevention of cross-infection are essential nursing responsibilities.

The WHO curriculum specifically includes infection prevention and control among the core patient-safety topics for health-professional education.

Literature related to communication and teamwork

Effective communication and teamwork are essential for preventing errors and ensuring continuity of patient care. Communication failures during handover, medication administration and clinical procedures can contribute to patient-safety incidents.

The WHO patient-safety curriculum identifies being an effective team player as a specific area of patient-safety education. It also emphasizes patient engagement and learning from errors.

Literature related to learning from errors and incident reporting

A positive safety culture encourages health professionals and students to identify hazards, report incidents and learn from mistakes rather than simply focusing on blame. Incident reporting can help health-care organizations identify system weaknesses and develop preventive measures.

The WHO curriculum includes "learning from errors to prevent harm" and "understanding and managing clinical risk" as core patient-safety topics.

Literature related to educational interventions

A 2025 systematic review examined the effectiveness of educational interventions for developing patient-safety competencies among undergraduate nursing students. The review highlights the importance of educational strategies specifically designed to improve patient-safety competencies rather than relying only on traditional classroom teaching.

This supports the use of approaches such as simulation, case-based learning, clinical demonstration, return-demonstration, structured feedback and supervised clinical practice.

Recent evidence on patient-safety competency

A 2026 analysis of 30 cross-sectional studies from 13 countries compared patient-safety competency among nurses and nursing students. It found that students generally reported higher competency scores than practicing nurses, but students also showed a gap between theoretical learning and practical application. Teamwork and responding to adverse events were among areas requiring attention. The authors recommended strengthening practical learning and interprofessional education.

Research gap

The reviewed literature indicates that:

- Patient safety should be incorporated into undergraduate nursing education.
- Students generally acquire patient-safety knowledge through classroom and clinical education.
- Clinical safety knowledge may be stronger than some broader safety-culture competencies[4].
- There may be a gap between theoretical knowledge and actual clinical performance.
- Simulation, supervised practice, feedback and competency-based assessment can strengthen patient-safety learning.
- More institutional-level studies are needed to identify the specific patient-safety learning needs of nursing students.

Therefore, the present study was undertaken to assess patient safety in clinical practices among 2nd-semester B.Sc. Nursing students of Regional College of Nursing, Guwahati, Assam, using a knowledge questionnaire together with recorded skill and practice scores.

Need for the present study

The present study is needed for the following purposes:

1. To assess the existing knowledge of B.Sc. Nursing students regarding patient safety.
2. To assess clinical skills and practices related to patient safety.
3. To identify gaps between theoretical knowledge and clinical performance.
4. To improve students' confidence and competency in providing safe patient care.
5. To provide baseline information that can be used for planning[5] future patient-safety education and training programmes.

Methodology

1. **Research approach:** Quantitative
2. **Research design:** Descriptive
3. **Study setting:** Regional College of Nursing, Guwahati, Assam
4. **Population:** 2nd-semester B.Sc. Nursing students
5. **Sample:** 2nd-semester B.Sc. Nursing students
6. **Sample size:** 42 students
7. **Sampling technique:** Consecutive sampling technique
8. **Inclusion criteria:** 2nd-semester B.Sc. Nursing students who were willing to participate in the study.
9. **Exclusion criteria:** 2nd-semester B.Sc. Nursing students who were absent or not willing to participate.
10. **Variables of the study:**
    a. **Study variables:** Patient safety in clinical practices; knowledge, skill and practice.
    b. **Demographic variables:** Age, previous clinical experience, duration of training, source of information, witnessed patient-safety incidents[3].
11. **Ethical considerations:**
    a. Permission was obtained from the concerned authority.
    b. Voluntary participation.
    c. Confidentiality and anonymity were maintained.

Schematic diagram of the research methodology

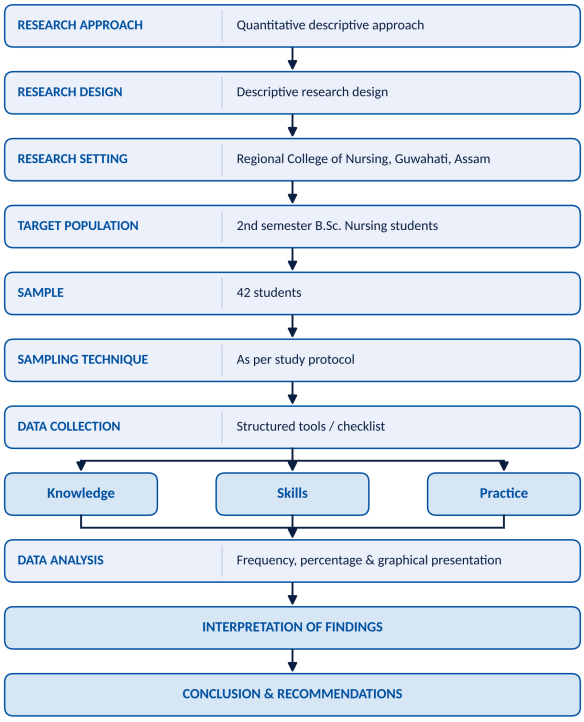


Figure 1. Schematic diagram of the research methodology

Data collection tools

1. Self-administered questionnaire
2. Interview schedule

Results

Analysis of demographic data

Table 1. Age distribution (N = 42)

Age group	Frequency (f)	Percentage (%)
18–19 years	12	28.6
20–21 years	23	54.8
22–23 years	6	14.3
24 years and above	1	2.4
Total	42	100

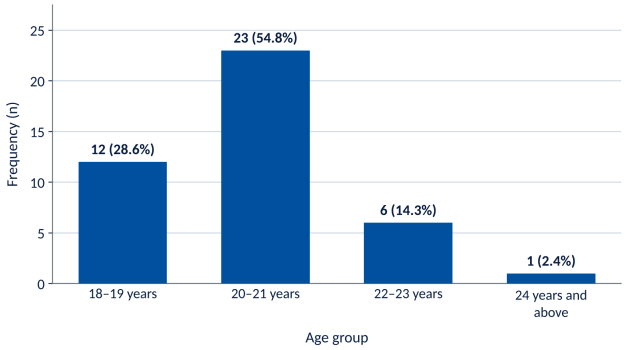


Figure 2. Distribution of students according to age (N = 42)

Interpretation: The majority of students, 23 (54.8%), were in the age group of 20–21 years. Twelve (28.6%) were aged 18–19 years, 6 (14.3%) were aged 22–23 years, and 1 (2.4%) was 24 years or above[6].

Table 2. Previous exposure to patient-safety training (N = 42)

Previous training	Frequency (f)	Percentage (%)
1 day	31	73.8
2 days	1	2.4
3 days	1	2.4
4 or more days	3	7.1
No previous training	6	14.3
Total	42	100

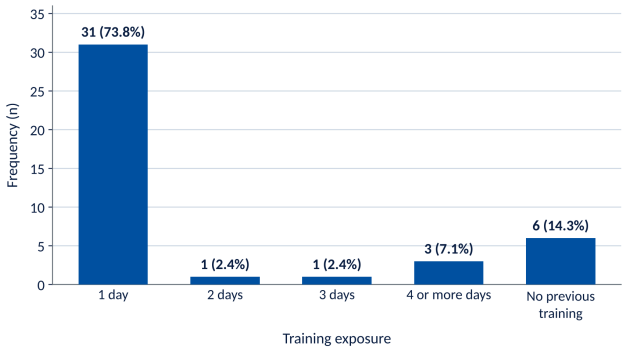


Figure 3. Previous exposure to patient-safety training (N = 42)

Interpretation: The majority, 31 (73.8%), had previously received 1 day of patient-safety training. Six (14.3%) had no previous training, while 3 (7.1%) had received training for 4 or more days. Only 1 (2.4%) student each had received 2-day and 3-day training.

Table 3. Source of information regarding patient safety (N = 42)

Source	Frequency (f)	Percentage (%)
Classroom teaching	32	76.2
Clinical posting	4	9.5
Health personnel	4	9.5
Books & internet	2	4.8
Total	42	100

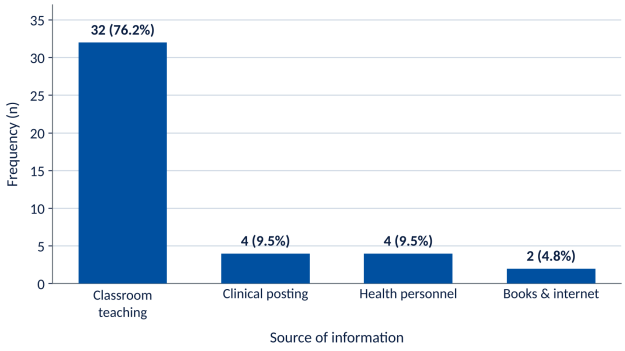


Figure 4. Source of information regarding patient safety (N = 42)

Interpretation: The major source of information regarding patient safety was classroom teaching, reported by 32 (76.2%) students. Clinical posting and health personnel were reported by 4 (9.5%) students each, while 2 (4.8%) students reported books and internet as their source.

Table 4. Previous witnessing of a patient-safety incident (N = 42)

Witnessed an incident	Frequency (f)	Percentage (%)
Yes	5	11.9
No	37	88.1
Total	42	100

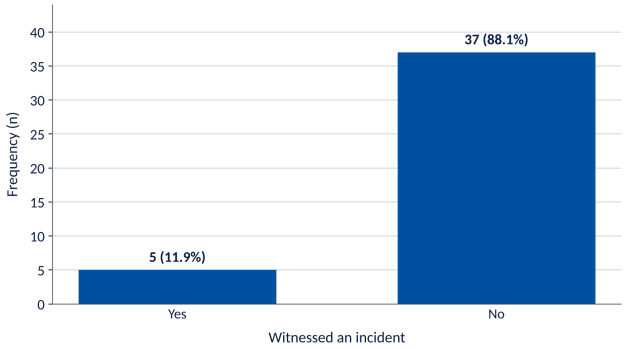


Figure 5. Previous witnessing of a patient-safety incident (N = 42)

Interpretation: The majority of students, 37 (88.1%), reported that they had not witnessed a patient-safety incident during clinical posting. Only 5 (11.9%) reported having witnessed an incident.

Overall demographic interpretation

Most participants were 20–21 years old (54.8%). A large majority had received one day of previous patient-safety training (73.8%), and classroom teaching was the predominant source of information (76.2%). Most students (88.1%) had not previously witnessed a patient-safety incident during clinical posting[7].

Data analysis for patient-safety scores

Table 5. Distribution of students according to level of knowledge regarding patient safety (N = 42)

Level of knowledge	Frequency (f)	Percentage (%)
Adequate	37	88.10
Moderate	1	2.38
Inadequate	4	9.52
Total	42	100.0

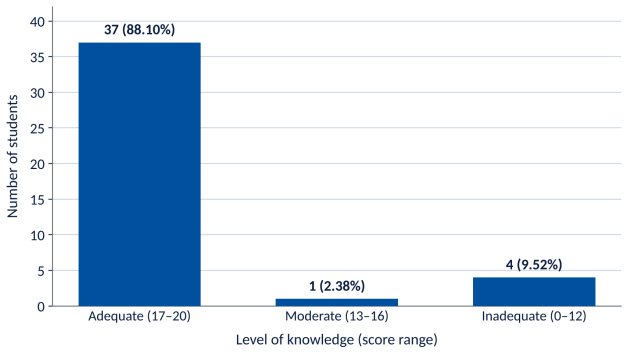


Figure 6. Distribution of students according to level of knowledge regarding patient safety (N = 42; knowledge score: adequate 17–20, moderate 13–16, inadequate 0–12)

Mean knowledge score = 17.14

Interpretation: The majority of students, 37 (88.10%), had adequate knowledge regarding patient safety. Only 1 (2.38%) student had moderate knowledge, while 4 (9.52%) students had inadequate knowledge.

The mean knowledge score was 17.14 out of 20, indicating an overall high level of knowledge.

Table 6. Distribution of students according to level of patient-safety skill (N = 42)

Level of skill	Frequency (f)	Percentage (%)
Adequate	13	30.95
Moderate	25	59.52
Inadequate	4	9.52
Total	42	100.0

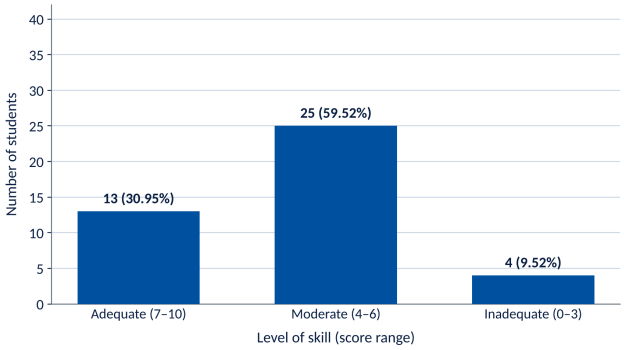


Figure 7. Distribution of students according to level of patient-safety skill (N = 42; skill score: adequate 7–10, moderate 4–6, inadequate 0–3)

Mean skill score = 5.74

Interpretation: More than half of the students, 25 (59.52%), demonstrated moderate skills related to patient safety. 13 (30.95%) had adequate skills, whereas 4 (9.52%) had inadequate skills.

The mean skill score was 5.74 out of 10, indicating that skills were predominantly at a moderate level.

Table 7. Distribution of students according to level of patient-safety practice (N = 42)

Level of practice	Frequency (f)	Percentage (%)
Adequate	2	4.76
Moderate	17	40.48
Inadequate	23	54.76
Total	42	100.0

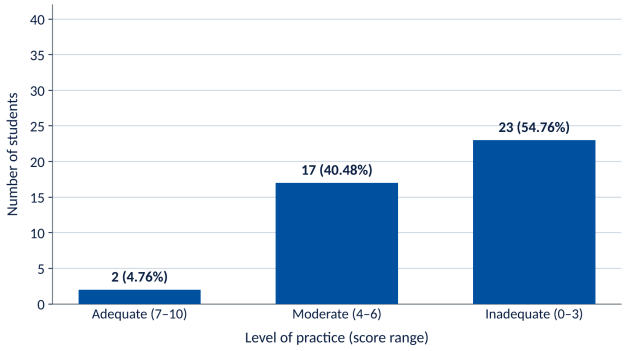


Figure 8. Distribution of students according to level of patient-safety practice (N = 42; practice score: adequate 7–10, moderate 4–6, inadequate 0–3)

Mean practice score = 3.57

Interpretation: The majority of students, 23 (54.76%), demonstrated inadequate patient-safety practice. 17 (40.48%) had moderate practice, while only 2 (4.76%) demonstrated adequate practice.

The mean practice score was 3.57 out of 10, indicating that patient-safety practices require considerable improvement.

Overall analysis

Table 8. Overall analysis of knowledge, skill and practice regarding patient safety (N = 42)

Domain	Mean score	Predominant level	Percentage (%)
Knowledge	17.14	Adequate	88.10
Skill	5.74	Moderate	59.52
Practice	3.57	Inadequate	54.76

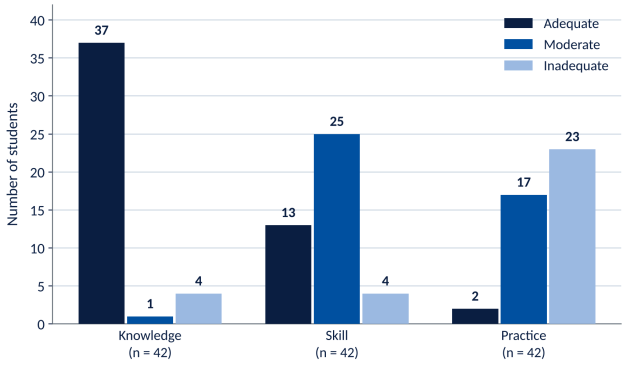


Figure 9. Level of patient safety (knowledge, skill and practice) among 42 B.Sc. Nursing students

Overall interpretation

Major finding: The data show a clear difference between the three domains:

- Knowledge → High (88.10% adequate)
- Skill → Mainly moderate (59.52% moderate)
- Practice → Mainly inadequate (54.76% inadequate)

This suggests that although the students possess good theoretical knowledge about patient safety, this knowledge is not being translated adequately into clinical skills and regular patient-safety practices.

Discussion

The present study was conducted to assess knowledge, skills, and practice regarding patient safety among 42 second-semester B.Sc. Nursing students of Regional College of Nursing, Guwahati, Assam.

The analysis showed that knowledge was the strongest dimension. Among the 42 students, 37 (88.10%) had adequate knowledge, 1 (2.38%) had moderate knowledge, and 4 (9.52%) had inadequate knowledge. This indicates that most students had a satisfactory theoretical understanding of patient-safety principles[8].

In contrast, the assessment of skills showed that 13 (30.95%) students had adequate skills, 25 (59.52%) had moderate skills, and 4 (9.52%) had inadequate skills. The predominance of moderate skill levels suggests that students may understand the principles of patient safety but require more opportunities to practice and demonstrate these skills under supervision.

The findings for practice were less satisfactory. Only 2 (4.76%) students demonstrated adequate practice, whereas 17 (40.48%) had moderate practice and 23 (54.76%) had inadequate practice. This indicates that the application of patient-safety principles in clinical practice is the major area requiring improvement.

The comparison of the three dimensions demonstrates a clear knowledge–skill–practice gap. Although 88.10% of students had adequate knowledge, only 30.95% had adequate skills and 4.76% had adequate practice. This suggests that theoretical knowledge alone does not guarantee appropriate clinical application.

The findings may be related to the students' limited clinical experience, developing confidence, insufficient opportunities for repeated practice, and the need for continuous clinical supervision. Therefore, nursing educators should emphasize demonstration, return demonstration, simulation-based learning, bedside teaching, supervised clinical practice, and regular feedback.

Overall, the analysis indicates that the students possess a good theoretical foundation in patient safety, but their clinical skills and especially their practical application need strengthening. Continuous patient-safety education and structured clinical supervision may help reduce the knowledge–practice gap and prepare students to provide safe and quality nursing care.

Summary

The present study was conducted to assess knowledge, skills, and practice regarding patient safety among 42 second-semester B.Sc. Nursing students of Regional College of Nursing, Guwahati, Assam. A quantitative descriptive approach was adopted, and data were collected using appropriate assessment tools covering the three dimensions of patient safety[9]. The collected data were analyzed using frequency, percentage, and mean scores and presented through tables and figures.

The findings showed that knowledge was the strongest domain. Out of 42 students, 37 (88.10%) had adequate knowledge, 1 (2.38%) had moderate knowledge, and 4 (9.52%) had inadequate knowledge. Regarding skills, 13 (30.95%) students had adequate skills, 25 (59.52%) had moderate skills, and 4 (9.52%) had inadequate skills. Regarding practice, only 2 (4.76%) students demonstrated adequate practice, 17 (40.48%) had moderate practice, and 23 (54.76%) had inadequate practice.

The overall findings demonstrate a clear knowledge–skill–practice gap. Although the majority of students possessed adequate theoretical knowledge, their clinical skills were predominantly moderate and their patient-safety practices were predominantly inadequate. This indicates that theoretical knowledge alone is not sufficient to ensure safe clinical practice[10].

Therefore, the study concludes that there is a need to strengthen the practical application of patient-safety principles among nursing students. Demonstration, return demonstration, simulation-based learning, supervised clinical practice, bedside teaching, regular observation, and continuous feedback should be incorporated into nursing education.

Overall, strengthening knowledge, clinical skills, and safe practices simultaneously can improve students' competence and confidence in patient care[11]. Early and continuous emphasis on patient safety during the B.Sc. Nursing programme can help students develop safe professional practices and contribute to quality nursing care, prevention of avoidable harm, and improved patient outcomes.

Conclusion

Hence, the investigator felt the need to conduct the study entitled "Patient safety in clinical practices among B.Sc. Nursing Students of Regional College of Nursing, Guwahati, Assam". The study may provide useful information about students' patient-safety knowledge and clinical performance and may help in improving patient-safety education and clinical supervision.

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REVIEW ARTICLE · HEALTH AND NURSING SCIENCES

NEUROPSYCHIATRIC DISORDERS: A COMPREHENSIVE OVERVIEW

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KEYWORDS

Anxiety, Neuropsychiatric Disorders, Neurodegenerative Diseases, Depression, Mood Disorders, Psychotic Disorders

ABSTRACT

Neuropsychiatric disorders pose a multifaceted and complicated challenge in present-day medicine and neuroscience. These complex disorders present a unique set of challenges in both clinical practice and research, as they display diverse symptoms and underlying causes. To successfully care for those affected, it is essential for healthcare professionals and researchers to comprehend the complexities of this field. This includes the different types of neuropsychiatric disorders, their etiologies and risk factors, as well as current approaches to diagnosis and management. The discussion starts by investigating the diverse categories of neuropsychiatric disorders, such as neurodegenerative conditions, mood disorders, psychotic disorders, traumatic brain injury, and stroke-related neuropsychiatric disorders. Each of these groups is characterized by particular clinical presentations, pathophysiology, and impact on cognitive, emotional, and behavioral functioning. The review then delves into the complicated interaction of genetic, environmental, neurochemical, and vascular factors that contribute to the development and progression of these disorders. With advances in diagnostic techniques, such as neuropsychological assessments, neuroimaging, genetic testing, and biomarker analysis, accurate identification and characterization of these conditions have significantly improved. Furthermore, the article discusses the various treatment and management strategies utilized, including pharmacological interventions, psychological therapies, neuromodulation procedures, and lifestyle modifications. By synthesizing the current understanding of this complex field, this review seeks to help healthcare professionals, researchers, and policymakers in improving the quality of life and outcomes for people affected by the vast and complex range of neuropsychiatric disorders.

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https://internationalresearchinsights.com/articles/neuropsychiatric-disorders-comprehensive-overview

Introduction

The complexity of neuropsychiatric disorders poses a significant challenge in the realm of contemporary medicine and neuroscience. Psychiatric symptoms are often brought on by neurological problems. It is advised to refer to psychological symptoms or syndromes brought on by organic brain illnesses as “neuropsychiatric symptoms” or “neuropsychiatric syndromes.” Neuropsychiatric disorders are the term used to describe cerebral disorders that result in psychiatric symptoms[1].

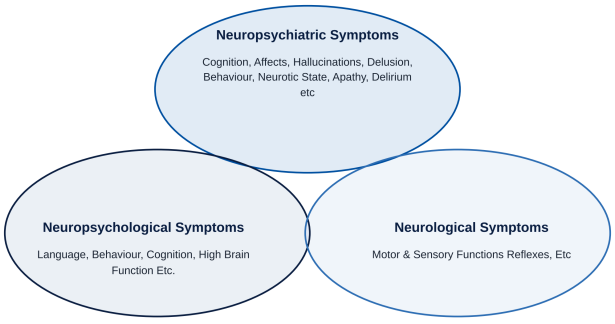


Figure 1. Structures of the clinical manifestations of cerebral disorders

Neuropsychiatric disorders are a subset of mental disorders for which there is some knowledge about the underlying brain pathology and, most importantly, this pathology is relevant to treatment. Neuropsychiatric

disorders are specific, clinically recognised conditions in which an individual's thoughts, perceptions, emotions, and/or overt behaviour cause suffering and interfere with the individual's daily life. By definition, these illnesses have a significant impact on how a person expresses their personality; they adversely affect comfort and willingness to interact with others, as well as attitudes towards oneself, mood, and disposition. Individual variations have traditionally been understood to reflect variability related to categorical distinctions[2]. It presents a diverse array of difficulties both in clinical practice and research, due to the wide range of symptoms and underlying causes these disorders can exhibit. A thorough grasp of this multifaceted field is crucial for healthcare professionals and researchers to provide effective and comprehensive care to those affected. This entails understanding the different types of neuropsychiatric disorders, their root causes, and the current methods of diagnosis and management. Our discussion delves into these topics by exploring the various classifications of neuropsychiatric disorders, such as neurodegenerative conditions, mood disorders, psychotic disorders, traumatic brain injury, and stroke-related neuropsychiatric disorders. Each subgroup is characterized by distinct clinical presentations, pathophysiologies, and impacts on cognitive, emotional, and behavioral functioning. Additionally, we investigate the intricate interplay of genetic, environmental, neurochemical, and vascular factors that contribute to the development and progression of these disorders. With advancements in diagnostic techniques like neuropsychological assessments, neuroimaging, genetic testing, and biomarker analysis, the ability to accurately identify and classify these conditions has improved significantly. Furthermore, this article examines various treatment and management strategies, including pharmacological interventions, psychological therapies, neuromodulation techniques, and lifestyle modifications. By synthesizing the current knowledge of this complex field, our review aims to inform and guide healthcare professionals, researchers, and policymakers in their efforts to enhance the quality of life and outcomes for those affected by the vast spectrum of neuropsychiatric disorders[3,4].

Literature Review

The study of neuropsychiatric disorders has changed a lot because of progress in neuroscience, psychiatry, pharmacology and clinical psychology. The existing information shows that these disorders happen because of interactions between genetic tendencies, brain problems, outside influences and social and psychological factors. The information that has been reviewed gives a base for understanding how to classify, diagnose, treat and manage these disorders over time.

Miyoshi, Morimura and Maeda (2010), in *Neuropsychiatric Disorders*, talked about the link between brain problems and mental symptoms. Their work gives an idea for understanding how changes in the brain's structure and functions can lead to problems with thinking, feeling and acting.

Berton and Nestler (2006) looked at new ways to create antidepressant drugs beyond the usual ideas about chemicals in the brain. Their work shows how important it is to look at brain processes and create better medicines for mood problems. Also, Drevets (2000) studied brain scans linked to mood disorders, showing how looking at brain structure and function can help understand health issues.

Studies about neurodegenerative diseases have also added a lot to the field. The Alzheimer's Association (2021) wrote about the impact of Alzheimer's disease on thinking, social life and health care. Research on Parkinson's disease and Huntington's disease also shows how the slow loss of brain cells can cause both brain symptoms and mental problems.

Merikangas et al. (2011) studied how common bipolar disorders are and what factors are connected to them. Kessler et al. (2005) looked at how common mental disorders are and when they usually start. These studies show how common mental health problems are and how important it is to find them early, diagnose them properly and start treatment quickly.

Lewis and Lieberman (2000) studied the way schizophrenia develops and what is happening in the brain. They pointed out the need to understand the underlying biology of this disorder. Karlsgodt et al. (2008) studied how changes in brain connections are linked to schizophrenia, supporting the idea of looking at brain networks instead of just focusing on single chemicals.

Geschwind and Flint (2015) looked at how genes and genetic research help in understanding disorders. Their work shows that genetic risks work with growing-up influences, not just by themselves, to create most mental health problems.

Najjar et al. (2013) studied how brain inflammation and mental illness are connected. Gorelick (2019) talked about inflammation and thinking problems. This information suggests that some brain inflammation might help cause mental health issues, but the exact role is different for each condition and needs more study.

Lezak et al. (2012) wrote an account about how to check brain function, making clear how important it is to test memory, focus, language and thinking skills. Research on behavior therapy by Hofmann et al. (2012) also shows how helpful evidence-based therapy is for many mental health problems.

Cramer et al. (2011) looked at the chance to use brain changes for treatment. This work helps explain how people can recover after brain injuries. Studies on care that includes areas of health also show how important it is to use medical, mental, recovery and social help together.

Overall, the current information offers a lot of knowledge about the body processes, signs, how to diagnose and how to treat disorders. However, the large number of these disorders and the unclear reasons behind them show that a full team-based look is needed.

Research Methodology

Research design

This study uses an analytical narrative review to look at neuropsychiatric disorders. It pulls together existing work on how neuropsychiatric disorders show up in patients, what causes them, how to test for them, how to treat them and what problems arise in care.

The study does not collect data, run experiments, talk to patients or run a fresh survey.

Sources of data

The research draws on sources, including:

- Peer-reviewed research articles and review papers.
- Standard textbooks on neuroscience, psychiatry, pharmacology and neuropsychology.
- Clinical and diagnostic publications.
- Publications by known health groups.
- Relevant science books on disorders, psychiatric conditions, brain scans, biomarkers and treatment methods.

Databases such as PubMed, PubMed Central, Google Scholar and other medical journal sites can be used to find papers.

Selection of literature

A paper is relevant if it talks about one or more of these themes: neurodegenerative diseases, mood disorders, anxiety disorders, psychotic disorders, brain injury, stroke-related neuropsychiatric conditions, genetic and environmental risk factors, tests for diagnosis and ways to treat.

Peer-reviewed work, well-known scientific books and recent evidence-based guides get priority. Older foundational works are also kept when they give theory or history.

Method of analysis

The chosen papers are read for themes. They are grouped into categories: disease types, causes, how the disease works, diagnosis, treatment and health care problems.

Findings from papers are compared to spot common ideas, where doctors agree or disagree and what still needs more study.

Methodological limitations

Because this is a narrative review, it does not promise a systematic search or a statistical meta-analysis. The conclusions depend on what papers exist, how good they are and how they fit together.

Research Gap

Although many studies have examined neuropsychiatric disorders, many gaps remain when we try to combine knowledge from neurology, psychiatry and drug science.

First, most existing literature focuses on conditions such as Alzheimer’s disease, Parkinson’s disease, depression or schizophrenia. It does not look at the range of neuropsychiatric disorders through one clear framework.

Second, the link between brain abnormalities and mental symptoms needs joint study. Cognitive decline, mood swings and behavior changes can all happen together. The ways they work, what they mean for patients and how to treat them are quite different.

Third, new tools like brain scans, gene tests, biomarker studies and brain-testing methods give ways to diagnose neuropsychiatric disorders. Their real use in clinics, how easy they are to get, how much they cost and how reliable they are still matter a lot.

Fourth, the success of drugs, talk therapy, brain-stimulation techniques and lifestyle changes depends on the kind of disorder and on each person. A full picture of how these options fit into treatment plans is still missing.

Fifth, differences in mental-health access, test availability, rehab centers and specialist care can affect outcomes in places with few resources.

Finally, we need synthesis to link new ideas about brain inflammation, neuroplasticity, genetic risks and environmental effects with real-world care for neuropsychiatric disorders.

The present study meets these gaps by looking at what we know about classification, causes, diagnosis, treatment and the wider impact of neuropsychiatric disorders.

Research Questions

The present study seeks to address the following research questions:

1. What are the major categories of neuropsychiatric disorders and how do their clinical manifestations differ?
2. What genetic, environmental, neurochemical, inflammatory and vascular factors contribute to the development and progression of these disorders?
3. How do neuropsychological assessments, neuroimaging, genetic testing and biomarker analysis contribute to the diagnosis and characterization of these conditions?
4. What are the respective roles of pharmacological interventions, psychological therapies, neuromodulation, rehabilitation and lifestyle modifications in their management?
5. What are the major challenges in the diagnosis, individualized treatment and long-term care of individuals affected by neuropsychiatric disorders?

Research Objectives

The principal objective of this study is to develop an integrated understanding of neuropsychiatric disorders through an examination of their biological foundations, clinical characteristics, diagnostic procedures and therapeutic approaches.

The specific objectives are:

1. To classify neuropsychiatric disorders and examine their characteristic cognitive, emotional, motor and behavioural manifestations.
2. To investigate the genetic, environmental, neurochemical, inflammatory and vascular factors associated with their development.
3. To evaluate the role and limitations of diagnostic techniques, including neuropsychological testing, neuroimaging, genetic assessment and biomarkers.
4. To examine pharmacological, psychological, neuromodulatory and lifestyle-based treatment strategies.
5. To assess the importance of healthcare, individualized treatment planning and rehabilitation in improving patient outcomes.
6. To identify existing knowledge gaps and suggest directions for future research, clinical practice and healthcare policy.

Hypotheses

Since this study is based on a narrative review rather than primary statistical research, the following propositions are framed as conceptual research hypotheses.

Primary hypothesis (H_1): An integrated understanding of the psychological and environmental dimensions of neuropsychiatric disorders supports more comprehensive diagnostic and management strategies than an approach focused on a single dimension.

Null hypothesis (H_0): An integrated understanding of the psychological and environmental dimensions of neuropsychiatric disorders does not provide additional value for diagnostic and management strategies compared with an approach focused on a single dimension.

Hypothesis: Early recognition, individualized treatment and multidisciplinary care are important components of the effective management of neuropsychiatric disorders.

These hypotheses guide the review and interpretation of existing literature. They are not statistically tested in the study because no primary dataset has been collected.

Scope, Significance and Relevance of the Study

Scope of the study

The scope of the study includes the psychiatric, pharmacological, psychological and rehabilitative aspects of neuropsychiatric disorders. It covers neurodegenerative diseases, mood disorders, anxiety disorders, psychotic disorders, traumatic brain injury and stroke-related neuropsychiatric conditions.

The study looks at the risk factors, the way the body works in these conditions, how they are diagnosed and how they are treated. It also looks at methods like advanced brain scans, genetic testing, biomarker analysis and changing brain activity.

The review focuses mainly on scientific papers. It does not try to find out how often each disorder happens in a group of people or compare how well treatments work using new experiments.

Significance of the study

Neuropsychiatric disorders can have a big impact on thinking, feelings, how the body works, relationships, jobs and how good life feels. These problems can also affect family members, people who take care of others, healthcare places and the whole community.

The importance of this study is that it tries to bring all the important information into one clear picture. By looking at both the brain and mental health parts, the research shows how similar symptoms can be and how to choose the right way to diagnose and treat them.

The study can also help find ways to spot problems early, reduce the shame around these issues, make rehabilitation

services stronger and encourage doctors and other health workers to work together.

Relevance of the study

The study is important for psychiatrists, neurologists, pharmacists, psychologists, neuropsychologists, people who help with recovery, researchers and students in medicine and health sciences.

For people in medicine, the discussion about how the brain chemicals work, the medicines used, side effects and how well treatments work is very important. The study also matters for health because getting the right mental health care and brain care is still a big problem in many areas.

International Perspective

Neuropsychiatric disorders are a problem all over the world because they affect people in different places, different income levels and different cultures. But the number of people who have these disorders, how they are seen, how easy it is to get treatment and how much they are affected by the disorder can be different from one group to another.

The World Health Organization gives a way to understand mental health as part of general health. Its work says that it is important to have care that is easy to get, focused on the person and that helps prevent problems, spot them early and protect the rights and dignity of people who are affected.

In the United States, research centers and medical groups have done a lot of work on brain diseases, how to diagnose health issues, brain scans and medicines. The Diagnostic and Statistical Manual of Mental Disorders made by the American Psychiatric Association is a way to list mental disorders. It is also important to think about the real situation and any brain problems that might be there.

In Europe, research has helped create treatments that are based on evidence, ways to help people recover from brain issues, and studies on brain inflammation and how parts of the brain connect. International teamwork has also helped study problems, how brain diseases work and what happens to people who have brain injuries over time.

In low- and middle-income countries like India, it can be hard to get help for neuropsychiatric disorders because there are not enough doctors, not everyone can get tests, money is a problem, there is shame and there are not enough ways to help people get better. These problems can make it hard to find out what is wrong and stop treatment.

An international view shows that just having advances is not enough. Good care also needs health systems, trained workers, ways that fit the culture, awareness in the public and a way to keep treatment going.

International information should be looked at carefully because the way health systems work, how people are tested, the people involved and the treatments available are very different in different countries.

The overall relevance of this study comes from its approach to understanding neuropsychiatric disorders. These disorders often cannot be fully explained by looking at one area—like brain function or mental health—on its own. Instead, they involve a mix of brain structure, brain chemicals, genetic risk, life experiences, mental processes and social factors. This means that a single view is usually not enough to get the picture.

For doctors and healthcare workers, this combined view can help them spot overlapping symptoms and decide which tests or treatments might be most useful. For people working in sciences, it offers a way to better understand how medicines work, how patients respond and what side effects may occur.

The study also matters for health. Early care, recovery support, teaching caregivers and making sure everyone has access to services all make a big difference in how people live with these conditions.

Research into things like biomarkers, the brain's ability to change, how parts of the brain connect and tailored treatments can help us understand these disorders better. This knowledge can lead to more focused treatments.

The main contribution of this review is bringing together science from different areas and showing how it can be used to provide care that truly focuses on the patient.

Results

The results below are based on a look at the existing research. They are not from an experiment or a survey with numbers.

Diversity of clinical manifestations

The studies show that neuropsychiatric disorders can show up in different ways. People might have problems with memory, feel sad or anxious, have hallucinations, struggle with movement or act differently. These symptoms vary a lot depending on the disorder and the person.

Multifactorial etiology

Experts agree that these disorders have multiple causes. Genes, brain chemistry, stress, inflammation in the brain, blood vessel health and early development can all play a role. No one cause explains everything. Different combinations of these factors lead to problems.

Importance of accurate diagnosis

Doctors rely on tests that look at thinking and behaviour as well as clinical interviews to find problems. Tools like brain scans, genetic tests and blood tests can help in some cases. These tools only work well when the evidence is strong and the situation fits.

Role of multimodal treatment

The research shows that treatment should match the disorder. Some people need medicine; others benefit from therapy, rehabilitation or changes in lifestyle. In some cases, techniques that change brain activity can help—under the care of specialists.

Need for individualized care

Differences in how the illness works, what symptoms a person has, how well they function and how they respond to treatment mean care must be personalized. In long-term or worsening illnesses, family members and teams of doctors often need to work together.

Continuing research needs

There are still questions to answer. We need to test biomarkers in real-world settings, design treatments for each person, track outcomes over time, understand how brain inflammation works, make sure everyone can access testing and see how well team-based care helps.

Types of Neuropsychiatric Disorders

Neurodegenerative disorders

Neurodegenerative disorders, such as Alzheimer's disease[5], Parkinson's disease[6], and Huntington's disease[7], are characterized by the progressive degeneration of specific neuronal populations, leading to a wide range of cognitive, motor, and behavioral symptoms. These disorders often present significant diagnostic and therapeutic challenges, as they involve complex pathological processes that are not yet fully understood.

- **Alzheimer's disease:** This is the most prevalent form of dementia, marked by the gradual deterioration of cognitive functions, such as memory, language, and executive abilities. Alzheimer's disease is associated with the accumulation of amyloid-beta and tau proteins in the brain, which contribute to neuronal death and synaptic dysfunction.
- **Parkinson's disease:** This disorder is primarily characterized by motor symptoms, including tremors, rigidity, and bradykinesia (slowness of movement), but it can also involve cognitive impairments and psychiatric disturbances, such as depression and anxiety. Parkinson's disease is linked to the degeneration of dopaminergic neurons in the substantia nigra region of the brain.
- **Huntington's disease:** Huntington's disease is an inherited disorder characterized by a triad of motor, cognitive, and psychiatric symptoms. It is caused by a genetic mutation that leads to the abnormal production of the huntingtin protein, resulting in the selective loss of neurons in various brain regions.

Mood disorders

Mood disorders, including major depressive disorder[8] and bipolar disorder[9], are characterized by persistent and pervasive disruptions in an individual's emotional state. These disorders can significantly impact an individual's ability to function in various domains, such as social, occupational, and personal. Anxiety disorders[10], which are closely linked to mood disorders, can further compound the challenges faced by individuals with neuropsychiatric conditions.

- **Major depressive disorder:** Major depressive disorder is a condition marked by persistent feelings of sadness, hopelessness, and anhedonia (loss of pleasure in activities), often accompanied by changes in sleep, appetite, and cognitive function.
- **Bipolar disorder:** Bipolar disorder is characterized by alternating episodes of mania (elevated mood and energy) and depression, which can significantly disrupt an individual's personal, social, and occupational life.
- **Anxiety disorders:** Anxiety disorders, such as generalized anxiety disorder, panic disorder, and social anxiety disorder, are characterized by persistent and excessive feelings of worry, fear, and apprehension, which can lead to physical symptoms and avoidance behaviors. Another significant neuropsychiatric disorder is anxiety, which encompasses a range of conditions characterized by persistent and excessive worrying, fear, and physiological arousal. Anxiety disorders, such as generalized anxiety disorder, social anxiety disorder, and panic disorder, have been linked to imbalances in the neurotransmitter systems responsible for regulating the stress response, including the gamma-aminobutyric acid (GABA) and noradrenergic systems[11]. Neuroimaging studies have consistently shown altered activity and connectivity in brain regions involved in threat detection and emotional processing, such as the amygdala and prefrontal cortex[12].

Psychotic disorders

Psychotic disorders, such as schizophrenia[8] and schizoaffective disorder[13], are characterized by disturbances in perception, cognition, and reality testing,

often manifesting in hallucinations, delusions, and disorganized thought patterns. These disorders can have a profound impact on an individual's social, occupational, and personal functioning, and are often associated with significant stigma and challenges in access to care.

- **Schizophrenia:** Schizophrenia, a chronic and debilitating neuropsychiatric disorder, is characterized by a range of symptoms, including hallucinations, delusions, disorganized speech and behavior, and cognitive impairments. The etiology of schizophrenia is multifaceted, involving genetic, environmental, and neurobiological factors. Disruptions in the dopaminergic and glutamatergic neurotransmitter systems, as well as structural and functional alterations in brain regions such as the prefrontal cortex, temporal lobe, and basal ganglia, have been implicated in the pathophysiology of schizophrenia[14,15].
- **Schizoaffective disorder:** Schizoaffective disorder involves a combination of psychotic symptoms, such as hallucinations and delusions, along with mood disturbances, such as periods of mania and depression.

Traumatic brain injury

Traumatic brain injury[16] can lead to a range of neuropsychiatric symptoms, including cognitive impairments, mood disturbances, and behavioral changes. The heterogeneous nature of traumatic brain injury and its complex pathophysiology make it a challenging condition to diagnose and manage effectively.

Stroke-related neuropsychiatric disorders

Stroke-related neuropsychiatric disorders[17] can arise as a consequence of the neurological insult, leading to a variety of cognitive, emotional, and behavioral changes. Understanding the specific mechanisms underlying these disorders and their impact on patient outcomes is an area of active research.

Underlying Causes and Risk Factors

Genetic factors[18]

The role of genetic factors is significant in the development of many neuropsychiatric disorders. Researchers have identified specific gene variants and polymorphisms that may increase susceptibility to these conditions.

Environmental factors[19]

In addition to genetic factors, environmental factors can also contribute to the emergence and progression of neuropsychiatric disorders. Exposure to toxins, traumatic life events, and adverse social determinants of health are all examples of environmental factors that may play a role.

Neurochemical imbalances[20]

Dysregulation of neurotransmitter systems, including serotonin, dopamine, and glutamate, has been linked to the development of various neuropsychiatric disorders. This has led to the development of targeted pharmacological interventions to address these imbalances.

Neuroinflammation[21]

Neuroinflammatory processes are increasingly recognized as playing a key role in the etiology and progression of numerous neuropsychiatric disorders. This highlights the potential for anti-inflammatory therapies to be used in their management.

Vascular factors[21]

Vascular factors, such as cerebrovascular disease and impaired cerebral blood flow, have been associated with the development of certain neuropsychiatric disorders. This emphasizes the importance of addressing cardiovascular health in the prevention and treatment of these conditions.

Diagnostic Approaches

By acknowledging the wide spectrum of neuropsychiatric disorders and their distinct traits, health professionals can improve their diagnostic and treatment strategies to effectively address the specific needs of individuals affected by these intricate conditions.

Neuropsychological assessments[23]

An all-encompassing evaluation of neuropsychological functioning is crucial in accurately diagnosing and describing neuropsychiatric disorders, as it offers valuable information about an individual's cognitive, emotional, and behavioral abilities.

Neuroimaging techniques[24]

Cutting-edge neuroimaging techniques, such as magnetic resonance imaging (MRI), positron emission tomography (PET), and functional MRI (fMRI), have significantly advanced our understanding of the structural and functional changes associated with various neuropsychiatric disorders.

Implementing genetic testing[25]

Genetic testing has gained importance in the diagnosis and risk assessment of specific neuropsychiatric disorders, enabling the detection of relevant genetic markers and offering potential for personalized treatment approaches.

Analyzing biomarkers[26]

The recognition and analysis of applicable biomarkers, including neurochemical, inflammatory, and metabolic markers, can provide valuable insights into the underlying

pathophysiology of neuropsychiatric disorders and aid in the development of more precise diagnostic tools.

Treatment and Management Strategies

Pharmacological interventions[27]

Interventions involving medication have long been the primary method of treating neuropsychiatric disorders, including the use of antidepressants, antipsychotics, and mood stabilizers. However, the quest for more targeted and effective medication options continues, driven by a deeper understanding of the underlying mechanisms of these disorders.

Psychological therapies[28]

Various forms of psychological therapies, such as cognitive-behavioral therapy, psychotherapy, and mindfulness-based interventions, have been proven effective in managing a variety of neuropsychiatric disorders. These therapies are often used in combination with pharmacological treatments.

Neuromodulation techniques[29]

Newly developed techniques in neuromodulation, such as transcranial magnetic stimulation (TMS) and deep brain stimulation (DBS), show promise in treating specific neuropsychiatric conditions. These techniques offer alternative options for individuals who do not respond well to traditional interventions.

Lifestyle modifications

Modifying lifestyle practices[30], such as incorporating regular exercise, making dietary changes, and implementing stress management strategies, are becoming increasingly recognized as valuable complements to the treatment of neuropsychiatric disorders. These lifestyle changes can positively impact neuroplasticity and overall brain health.

Multidisciplinary approach

A comprehensive and individualized approach[31], involving healthcare professionals from various specialties, such as neurology, psychiatry, neuropsychology, and rehabilitation, is often necessary to provide effective care for individuals with neuropsychiatric disorders. This multidisciplinary approach ensures that all aspects of the disorder are addressed and treated appropriately.

Final Research Result

The overall result of the literature review is that neuropsychiatric disorders constitute a group of conditions involving interactions between neurological functioning, psychological processes and environmental influences.

The reviewed evidence supports the importance of combining clinical assessment with appropriate diagnostic investigations and disorder-specific treatment. Pharmacological treatment remains important for many conditions, while psychological interventions, rehabilitation, lifestyle measures and social support contribute to management where appropriate.

The study also indicates that emerging technologies and advances in neuroscience offer opportunities to improve clinical understanding. However, such advances should be incorporated into practice only when supported by evidence and interpreted within the broader clinical context.

The central conclusion is that comprehensive neuropsychiatric care requires an evidence-based and multidisciplinary approach. Further clinical research is needed to determine which combinations of therapeutic strategies produce the best outcomes for specific disorders and patient populations.

Findings of the Study

The major findings derived from the reviewed literature are as follows:

1. **Heterogeneous disease patterns:** Neuropsychiatric disorders differ considerably in their presentation, progression and functional consequences.
2. **Interconnected mechanisms:** Genetic, environmental, neurochemical, inflammatory and vascular factors may contribute to disease development, although their relative importance differs across conditions.
3. **Importance of recognition:** Timely identification of emotional, behavioural and neurological changes can support appropriate assessment and treatment planning.
4. **Clinical value of tools:** Neuropsychological assessments and neurological examinations remain important, while imaging, genetic testing and biomarkers offer additional information in selected cases.
5. **Need for treatment:** No single medication, psychological intervention or rehabilitation programme is suitable for every disorder.
6. **Value of care:** Coordinated treatment involving relevant medical and mental health professionals can address multiple dimensions of a patient's condition.
7. **Importance of rehabilitation:** Cognitive rehabilitation, functional support and psychosocial interventions are particularly relevant to individuals experiencing injury or progressive functional impairment.

8. **Continuing scientific uncertainty:** The precise mechanisms underlying neuropsychiatric conditions remain incompletely understood, and findings from biological research do not always translate directly into clinical benefit.
9. **Healthcare access challenges:** Stigma, constraints, limited specialist availability and unequal access to diagnostic services may impede timely and continuous care.
10. **Need for research:** Further well-designed clinical studies are necessary to evaluate long-term treatment outcomes, validate emerging biomarkers and establish the comparative effectiveness of integrated interventions.

These findings represent themes identified from the literature rather than independently measured outcomes or statistically verified relationships.

Conclusion

The understanding of neuropsychiatric disorders is imperative as they are intricate and multifaceted illnesses that require a thorough comprehension of their underlying mechanisms, diagnostic methods, and up-to-date advancements in treatment and management. These disorders cover a broad range of conditions that present significant challenges for individuals, families, and healthcare systems. Through collaborative efforts across various disciplines, healthcare professionals and researchers can contribute their expertise to enhance patient outcomes, quality of life, and pave the way for more effective interventions for those affected by these incapacitating disorders. As research continues to unravel the complex neurobiological mechanisms, improve diagnostic tools, and develop more efficient treatment strategies, it is crucial in enhancing the well-being of those impacted by these intricate and disabling disorders.

Neuropsychiatric disorders represent a diverse group of conditions that affect neurological functioning, cognition, emotions, behaviour and overall quality of life. Their underlying mechanisms involve varying combinations of susceptibility, neurochemical abnormalities, environmental influences, inflammatory processes, vascular factors and changes in brain structure and function.

The present review demonstrates the importance of understanding these disorders through an integrated clinical framework. Although substantial progress has been achieved in neuroimaging, neuropsychological assessment, pharmacological treatment, psychotherapy and rehabilitation, important challenges remain in diagnosis, treatment selection, long-term management and access to appropriate healthcare.

A comprehensive approach requires the selection of methods according to individual clinical needs and the use of evidence-based treatments suited to the specific disorder. Pharmacological interventions, psychological therapies, rehabilitation and lifestyle modifications may complement one another. Their application must be guided by clinical evidence rather than a uniform treatment model.

Future research should prioritize the validation of biomarkers, investigation of disease mechanisms, development of interventions and evaluation of multidisciplinary care. Greater investment in health awareness, professional training, rehabilitation facilities and equitable healthcare access is also necessary.

Ultimately, improving outcomes for individuals with neuropsychiatric disorders requires sustained collaboration among researchers, clinicians, pharmacists, psychologists, rehabilitation specialists, policymakers, patients and caregivers. Such collaboration can contribute to clinical management, improved quality of life and a more comprehensive understanding of the relationship between neurological and psychiatric health.

Suggestions

Based on the literature reviewed, the following suggestions are proposed:

1. **Promote screening:** Appropriate screening and clinical assessment should be encouraged when individuals exhibit persistent cognitive, emotional, behavioural or neurological changes.
2. **Strengthen collaboration:** Neurologists, psychiatrists, pharmacists, psychologists and rehabilitation professionals should coordinate care according to individual needs.
3. **Improve capacity:** Healthcare institutions should strengthen access to neuropsychological assessments, appropriate neuroimaging and relevant laboratory investigations.
4. **Encourage treatment:** Treatment plans should consider diagnosis, how bad the symptoms are, medical history, how well treatment works, side effects and what the person likes.
5. **Expand psychological care:** More people should be able to get cognitive behaviour therapy and other suitable psychological help.
6. **Strengthen rehabilitation services:** Cognitive, physical, job-related and social rehab should be added to care when needed.

7. **Support biomarker research:** New biomarkers and genetic tests must be checked for their real clinical use before they are used regularly.
8. **Increase awareness:** Education programs should fight stigma, teach health basics and push people to get the right professional help.
9. **Improve caregiver support:** Families and caregivers must get the education, handy tips and help services, especially when caring for people with worsening conditions.
10. **Promote clinical research:** More studies must look at long-term results, how well treatment works, how much it costs and what stops people from getting care in all kinds of groups.

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